Optimal Defense Spending and Taxation in a Dynamic Economy

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October 2026

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Abstract

This paper studies optimal defense spending and tax structure in a dynamic economy in which public spending protects productive capital. When capital income can be taxed to finance protection, productive capital plays a triple role in the economy: it is a productive asset, the object of public protection, and a tax base that can be used to finance its own protection. I show that an economy can adapt to higher destruction risk by reallocating adequate resources towards both protection and production, and capital and output don't fall monotonically, although welfare declines. When defense must be financed with distortionary taxes, a Ramsey government relies mainly on labor taxation and keeps long-run capital taxation near zero to preserve the productive base; during the rebuilding transition it may subsidize capital accumulation, and the subsidy is larger when destruction exposure is higher or defense is less effective. By contrast, a discretionary government chooses to tax installed capital to finance defense, even though that same capital requires protection. A *protection channel* amplifies the standard capital-tax distortion and generates a consumption-equivalent welfare loss of approximately 2.5 percent including the transition in the benchmark calibration. The results highlight the importance of fiscal credibility in an economy's ability to adapt to destruction risk. Using the Geopolitical Risk Index to discipline the stochastic threat process, the model reproduces the positive response of defense spending to risk innovations but predicts a more immediate and sharper response than observed in the data.

JEL Codes: E62, H21, H56

Keywords: optimal taxation, defense spending, commitment, geopolitical risk

*Department of Economics, Georgetown University. Email: jw2035@georgetown.edu. I am deeply grateful to my advisors, Toshihiko Mukoyama and Axel Anderson, for their guidance and support. I thank Behzad Diba, Mark Huggett, Joshua Teitelbaum, Margit Reischer, Xavier Gabaix, Vincent Sterk, Vincenzo Quadrini, Zhengyang Jiang, Chris Sleet, Domenico Ferraro, Dongho Song and Conor Walsh for their helpful comments. All errors are my own.

1 Introduction

Geopolitical risks fluctuate dramatically over time and can pose a major challenge for both defense spending and fiscal policy. In 2025, NATO members committed to raise defense and security-related spending to 5 percent of GDP by 2035, including spending to protect critical infrastructure (North Atlantic Treaty Organization, 2025). The Geopolitical Risk Index developed by Caldara and Iacoviello (2022) documents large and persistent changes in the geopolitical threats faced by individual countries. This raises a natural question of how governments should optimally respond to such fluctuating threat levels. Geopolitical risk can affect the economy through many channels: lower investment, reduced economic activity, military buildups and slower output growth. This paper focuses on a particular channel: the risk to installed capital. Historical evidence suggests that interstate conflicts can destroy as much as 25 percent of capital stock, as in Luetticke et al. (2025). Recent conflicts also illustrate how missile and drone attacks can damage energy facilities, infrastructure, and other industrial assets in conflict-affected areas. How should the government optimally spend to protect installed capital and preserve productive capacity? Defense can mitigate these losses, but it also competes for resources with productive investment, which accumulates capital. Financing protection through distortionary taxation complicates the problem. When capital income can be taxed to finance protection, capital plays a triple role in the economy: a productive asset, the object of public protection and a tax base that can be used to finance that protection.

This leads to the main questions of the paper. First, in the face of risks that can destroy productive capital, how should the government allocate resources between productive investment and protection, when both contribute to future productive capacity? Second, how should the government finance the protection when the same capital stock that benefits from protection can also be taxed to pay for it? Third, how does the inability to commit to future fiscal policy affect defense provision, investment and welfare?

To answer these questions, I develop a dynamic general equilibrium model in which the economy faces a stochastic geopolitical threat. Uncertainty is resolved sequentially in two stages. First, the exogenous geopolitical threat level evolves stochastically and determines the probability of a destructive event. Second, conditional on the observed threat state, the destructive event either occurs or does not occur. The government chooses the level of defense spending after observing the threat state but before the destruction state is realized. If the destructive event occurs, it destroys a fraction of the economy's capital stock, and this fraction decreases with the level of defense spending. Defense competes for resources with productive investment, which accumulates the capital stock. Thus, both defense and productive investment affect the evolution of productive capital, but through different margins. I first characterize the Pareto-efficient allocation as a benchmark.

I then turn to a decentralized economy in which defense is financed through distortionary labor and capital income taxes. I compare two types of government: a Ramsey government that can commit to future defense expenditures and tax rates, and a discretionary government that can

choose to re-optimize fiscal policy each period after observing the state. This comparison allows me to study how the government's ability to commit affects the provision of defense and the accumulation of productive capital.

This model generates the following key properties and findings. First, at a theoretical level, there is a threshold below which it is optimal to provide no defense at all, and once the threshold is crossed, optimal defense spending increases in the threat level. The zero defense threshold arises because the benefit from the first unit of defense may not justify its resource cost. This threshold is higher when a smaller stock of productive capital is at stake, when destruction risk is less severe, or when the protection technology is less effective. Using distortionary taxation to finance defense further raises this threshold and therefore expands the zero-defense, or inaction, region.

Second, the economy generates an endogenous adaptation to greater destruction risk. Greater destruction risk does not necessarily imply monotonically lower long-run capital and output. This property originates from the complementarity between defense and investment. Greater defense raises the destruction-adjusted return to investment, while greater investment expands the stock of capital worth protecting and makes defense more valuable. If the protection technology is sufficiently effective, greater destruction risk can induce the planner to reallocate more resources towards both defense and productive investment. This adaptation can partially offset the destruction of productive capital and the resulting decline in output. This adaptation is costly, however: it requires greater defense expenditure, labor supply and lower consumption, so welfare falls.

Third, the strength of this adaptation response depends importantly on fiscal policy and commitment. When defense must be financed by distortionary taxes, a Ramsey government chooses to subsidize capital during the rebuilding transition and relies primarily on labor taxation in the long run to preserve the productive base. A Ramsey government can partially reproduce the first-best adaptation response, although capital, output, and defense provision remain below their first-best levels. In contrast, a discretionary government taxes installed capital to finance defense, even though that same capital requires protection. This amplifies the standard capital-tax distortion emphasized by Klein et al. (2008): anticipated taxation reduces household saving and shrinks the capital base. A smaller capital base, in turn, lowers the value of protection and weakens defense, exposing the remaining capital to greater destruction and further reducing productive capacity. As a result, adaptation is substantially weaker under discretion. Quantitatively, the inability to commit leads to lower investment and defense provision and generates a consumption-equivalent welfare loss of 2.5 percent relative to Ramsey policies in the benchmark calibration.

Lastly, I use the Geopolitical Risk Index by Caldara and Iacoviello (2022) to discipline the evolution of geopolitical threat. Applying the first-best and Ramsey policy rules to common stochastic histories, I study how productive investment and defense respond to time-varying geopolitical risk. The model reproduces the positive response of defense spending to risk innovations, although the response is more immediate and sharper than in the data, consistent with the absence of adjustment frictions.

This paper illustrates the mechanism in the context of destructive risk arising from geopolitical threat and optimal defense provision, but this framework can be applied more broadly in environ-

ments in which productive assets face destructive risk and protection is an endogenous policy choice. Examples include natural-disaster mitigation against hurricanes, floods, wildfires, climate-related coastal risk, and cybersecurity.

Literature. This paper is related to several strands of literature. First, it contributes to the growing literature on geopolitical risk, conflict, and defense provision. A closely related paper is Valaitis and Villa (2026), who study the optimal financing of defense when defense capital reduces war risk and insures against wartime fiscal needs. They allow for debt and have only labor tax in their model. They find optimal policy relies more on debt and delays labor taxation. My paper focuses on a different margin and is complementary: productive private capital is both exposed to destruction and used as a tax base, linking defense provision to capital taxation, investment, and government commitment. A growing empirical literature studies the macroeconomic consequences of geopolitical risk and armed conflict. Federle et al. (2026), using a long-run cross-country dataset on wars, document large and persistent declines in output, capital stocks, and total factor productivity in countries where wars are fought. These findings are particularly relevant for the mechanism in this paper. Here, the investment response to risk is not necessarily monotone, and movements in the capital stock reflect the interaction of taxation, investment, protection, and destruction.

Related theoretical work studies the interaction between geopolitical competition and macroeconomic and financial outcomes. For example, Pflueger and Yared (2024) develop a dynamic model in which military spending, geopolitical risk, sovereign bond prices, and international hegemony are jointly determined. In contrast, this paper takes geopolitical risk as an exogenous stochastic state and focuses on how households and governments adjust productive investment, defense provision, and taxation when productive capital is exposed to destruction.

Second, the paper also builds on the Ramsey literature on optimal capital taxation and commitment. Classic results in Judd (1985) and Chamley (1986) characterize optimal long-run capital taxation, while Chari et al. (1994) emphasize the distinction between ex-ante and ex-post taxation of capital. The inability to commit not to exploit predetermined private decisions is a central time-consistency problem (Kydland and Prescott, 1977). A subsequent literature studies fiscal policy without commitment, including Benhabib and Rustichini (1997), Phelan and Stacchetti (2001), and Klein et al. (2008).

The contribution here is that this familiar commitment problem interacts with public protection. Anticipated capital taxation reduces saving and shrinks the future capital stock, as in the standard mechanism. But when capital is also the object of protection, a smaller capital stock lowers the value of defense, which can reduce protection and expose the remaining capital to greater destruction. Capital taxation therefore creates an additional *protection channel* that amplifies the standard capital-tax distortion.

Lastly, this speaks to the literature on disaster prevention and adaptation. Motoyama (2017) studies publicly financed disaster prevention when disasters destroy capital, while Fried (2022) and Hong et al. (2023) study adaptation to climate-related destruction of productive assets. Related work such as Cai and Lontzek (2019) and Barrage (2020) incorporates uncertainty and fiscal costs

into dynamic climate policy. In this paper, defense can be interpreted as a form of protective investment, similar to the adaptation literature. The contribution of this paper is to further ask how protection should be optimally financed in the presence of stochastic destruction, and how the lack of government commitment complicates this problem and can potentially generate additional welfare losses. This issue is particularly important for public expenditures whose effectiveness depends on government credibility.

I organize the paper as follows. Section 2 provides the empirical background. Section 3 introduces the model. Sections 4 and 5 analyze the deterministic model and compare policies and allocations under three regimes: the first best, Ramsey commitment, and discretion. Section 6 studies how allocations and policies respond to time-varying geopolitical risk under the first-best and Ramsey. Section 7 discusses the limitations of the model and concludes.

2 Data and Empirical Motivation

This section documents several empirical patterns that motivate the model. First, geopolitical threat varies substantially over time, and defense expenditure responds to these changes. This suggests that defense is an endogenous and state-contingent policy choice. Second, when geopolitical risks are realized in the form of armed conflict, they can cause substantial destruction of productive physical capital. Together, these facts motivate a model in which geopolitical threats generate a demand for defense and defense spending protects productive capital from destruction.

2.1 Geopolitical Risk and Defense Spending

Geopolitical Risk is defined by Caldara and Iacoviello (2022) as the threat, realization, and escalation of adverse events associated with wars, terrorism, and tensions among states and political actors that affect the peaceful course of international relations. Their Geopolitical Risk Index (GPR Index) is a monthly newspaper-based measure constructed from the share of articles discussing adverse geopolitical events and threats, using keyword searches related to war, military buildups, nuclear threats, terrorism, and conflict escalation. Figure 1 shows the aggregate GPR Index since 1900. Large increases in the index are associated with major episodes of geopolitical tension and conflict.

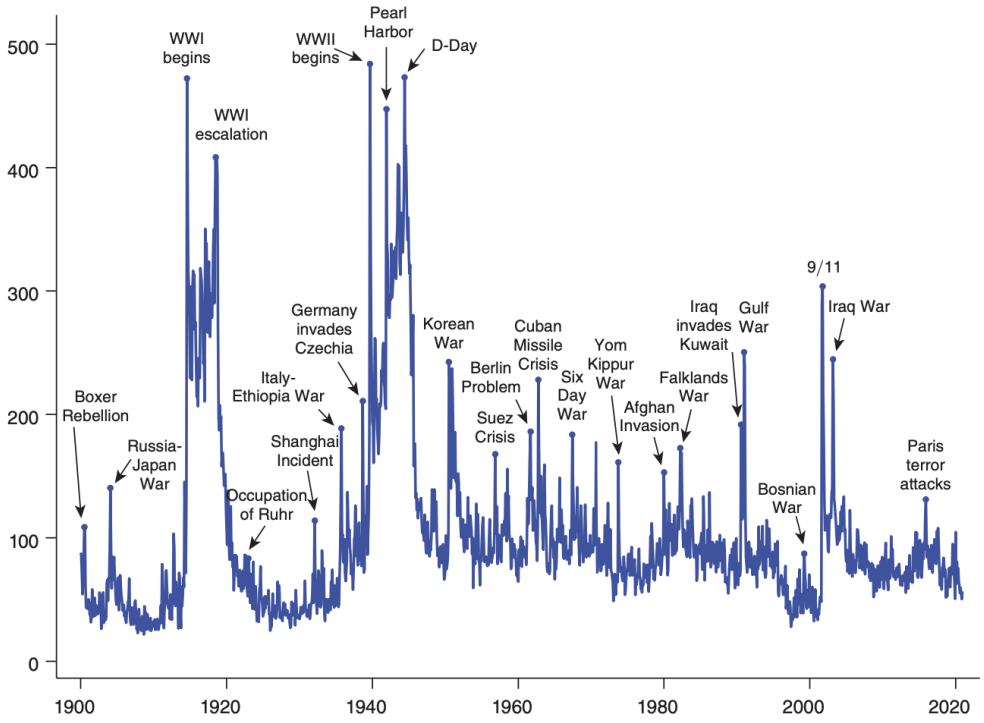


Figure 1: Aggregate Geopolitical Risk Index from 1900

The variation in geopolitical threat is often followed by changes in defense expenditure. To examine the relationship, I focus on the U.S.-specific GPR index and U.S. military spending. In particular, I focus on the relationship between the GPR index and U.S. military procurement, which comprises government purchases of goods and services from private contractors. Procurement is a useful measure because it captures the government's direct defense related adjustment in response to geopolitical risks. Data also shows that although military contracts account for less than half of total defense expenditure in my sample, they account for a large share of its variation.

Figure 2 shows the comovement between U.S military purchases and the U.S GPR index. In the U.S., heightened geopolitical risk is followed by an increase in defense-related spending.

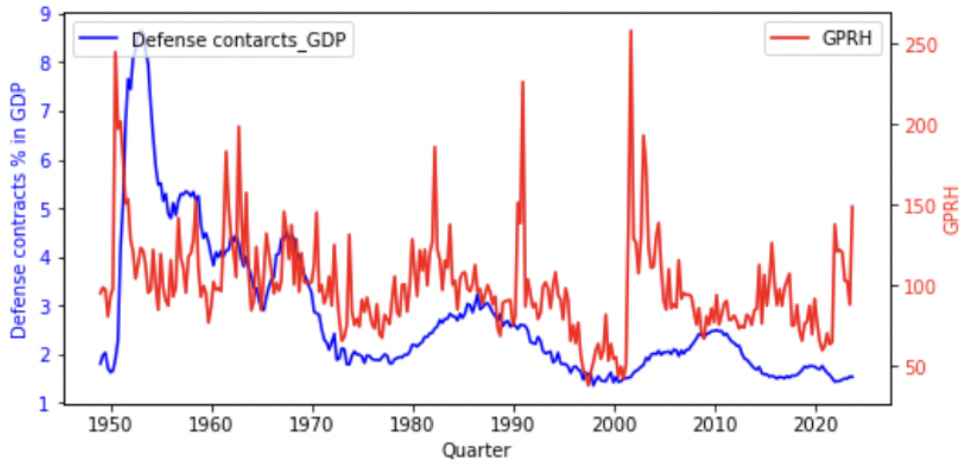


Figure 2: Co-movement of Military Purchases and Geopolitical Risk

Notes: This figure shows the comovement between the GPR index and military purchases. The GPR index is from Caldara and Iacoviello (2022), and military purchases are obtained from USAspending.gov.

To characterize the dynamic relationship between geopolitical risk and military procurement, I estimate a bivariate VAR using the GPR index and military purchases as a share of GDP. The impulse responses show that a positive innovation to geopolitical risk is followed by a persistent increase in military purchases. A one-standard-deviation innovation to the GPR index is followed by an increase of approximately 0.25 percentage points in military purchases as a share of GDP at its peak, about two years after the shock. This corresponds to roughly $50 billion in annual military purchases. Figure 3 reports the impulse responses.

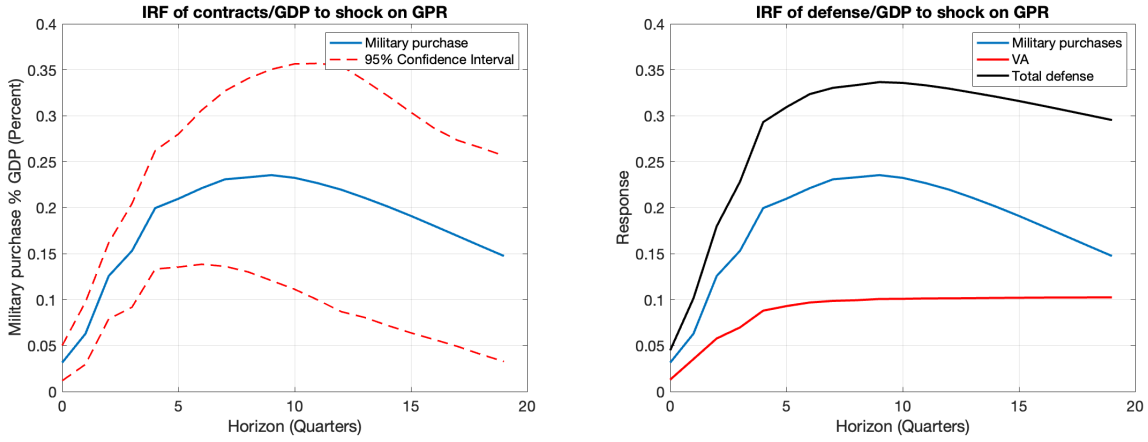


Figure 3: Response of Military Purchases to Geopolitical Risk

The evidence above shows that when geopolitical risk rises, governments devote additional resources to defense, but it does not tell us whether the observed response is optimal. Higher threat increases the potential benefit from protection, while the resources devoted to defense have an opportunity cost.

2.2 Conflict and Physical Capital Destruction

Armed conflict can destroy or disable industrial structures, machinery, energy infrastructure, transportation networks, housing, and commercial assets. These losses reduce the productive capital stock available to the economy after the conflict.

Evidence from recent conflicts illustrates the magnitude of these losses. World Bank (2022) estimates that in Ukraine, war-related damage was at roughly 10 percent of the capital stock exposed to conflict, with approximately 15 percent of exposed infrastructure damaged or destroyed. Damage was concentrated in transport, energy, housing, and commercial and industrial assets. Similar patterns are observed in other conflicts. World Bank (2025) estimates that more than a decade of conflict damaged nearly one-third of Syria's pre-war gross capital stock.

These facts motivate the second part of the model. I distinguish between the threat of conflict and the destruction that occurs if conflict is realized. Geopolitical threat determines the probability of entering the conflict state, while defense spending determines how much productive capital survives conditional on conflict. Defense therefore does not eliminate the underlying geopolitical threat in the model; instead, it reduces the fraction of exposed capital that is lost.

This setup also creates the fiscal tradeoff studied in the paper. Protecting productive capital requires current resources, but raising those resources through distortionary taxation can itself affect investment and capital accumulation. The next section develops a model in which the government chooses how much output to devote to protection and how to finance that protection through labor and capital taxation.

3 Model

The paper first introduces the stochastic environment, which speaks directly to evolving geopolitical risk. I then study a deterministic special case in which the destruction state occurs with certainty to isolate the main economic mechanism and derive comparative statics. Finally, I return to the stochastic environment for the quantitative analysis.

3.1 Environment

Time is discrete and infinite, $t = 0, 1, 2, \ldots$. The economy consists of a representative household, a representative competitive firm, and a benevolent government. The economy is subject to stochastic geopolitical risk. In each period, a war state may be realized, in which case a fraction of the productive capital stock is destroyed. The government can reduce capital destruction through defense expenditure, financed by taxes on labor and capital income.

Household preferences. The economy is populated by a unit measure of identical, infinitely lived households. The representative household has preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t, n_t), \qquad \beta \in (0,1),$$

where $c_t > 0$ denotes consumption and $n_t \in (0,1)$ denotes labor supply. The period utility function satisfies

$$u_c > 0, \qquad u_{cc} < 0, \qquad u_n < 0, \qquad u_{nn} < 0.$$

Production technology. Output is produced using capital and labor according to

$$y_t = F(K_t, n_t), \tag{1}$$

where K_t is the productive capital stock available at the beginning of period t. F is twice continuously differentiable, increasing and concave in both inputs, and exhibits constant returns to scale.

External threat and war risk. The economy faces geopolitical risk and the realization of risk can destroy capital stock. The external threat state z_t follows

$$z_{t+1} = \rho_z z_t + \epsilon_{t+1}, \tag{2}$$

where the innovation takes two values,

$$\epsilon_{t+1} \in \{-\Delta_z, +\Delta_z\}. \tag{3}$$

$j \in \{H, L\}$ denotes the realization of the innovation, with probability π_j. The threat state is mapped into the probability of war $\varphi_t \in [0,1]$ in period t, according to

$$\varphi_t = \Lambda(\eta_0 + z_t) = \frac{1}{1 + \exp[-(\eta_0 + z_t)]}. \tag{4}$$

Thus,

$$\Pr(s_t = W \mid z_t) = \varphi_t, \qquad \Pr(s_t = P \mid z_t) = 1 - \varphi_t. \tag{5}$$

The parameter η_0 determines the probability of war at the average level of geopolitical threat.

So there are two stages of uncertainty in the economy: threat uncertainty and event uncertainty. At the beginning of period t, an innovation to geopolitical threat determines the current threat state z_t and hence the probability of war φ_t. Conditional on φ_t, the peace or war outcome $s_t \in \{P, W\}$ is realized at the end of the period.

I further assume that, conditional on the current threat state, the realization of war and the innovation to next period's geopolitical risk are independent. Thus, from any current state, the continuation outcome combines the current peace or war realization with either a high or low threat innovation for the next period.

Conditional on the initial state (K_0, φ_0), let

$$h^t = (s_0, j_1, s_1, j_2, \ldots, s_{t-1}, j_t),$$

where $s_t \in \{P, W\}$ is the destruction realization during period t and $j_t \in \{H, L\}$ denotes the threat innovation entering period t. The history h^t, together with the initial state, determines the current state (K_t, φ_t).

Destruction and defense technology. Let x_t denote productive investment and d_t defense expenditure. Define the amount of capital exposed to destruction after investment as

$$A_t \equiv (1 - \delta)K_t + x_t, \tag{6}$$

where $\delta \in (0, 1)$ is the depreciation rate.

At the end of period t, the economy can be in either a peace state, $s_t = P$, or a war state, $s_t = W$. If peace is realized, no destruction happens and

$$K_{t+1} = A_t. \tag{7}$$

If war is realized, only a fraction $S(d_t)$ of exposed capital survives:

$$K_{t+1} = S(d_t)A_t. \tag{8}$$

The survival technology satisfies

$$S'(d) > 0, \qquad S''(d) \leq 0, \qquad S(d) \in [0, 1]. \tag{9}$$

Thus, greater defense expenditure increases the fraction of productive capital that survives a war, subject to diminishing returns.

This specification is natural when defense primarily protects a fixed geographic or strategic perimeter. For example, expenditure that secures a country's borders or hardens critical infrastructure may protect the asset within that perimeter. This type of protection doesn't require defense expenditure to increase proportionally with the value of those assets. Appendix B considers an alternative defense technology in which protection depends on defense expenditure relative to the value of the assets being protected, and examines how allocations and policies adjust under this specification.

I abstract from the potential deterrence effect of defense expenditure on the probability of conflicts. Such deterrence may be important, but the relationship between defense and the probability of conflict is itself an equilibrium outcome that also depends on the behavior of other strategic actors. Modeling this strategic interaction is beyond the scope of this paper.

For the analytical and quantitative exercises, I use the functional form

$$S(d) = 1 - ae^{-bd}, \tag{10}$$

where a governs the fraction of exposed capital destroyed in the absence of defense, so $S(0) = 1 - a$, and b governs the effectiveness of defense expenditure in reducing destruction.

Government budget constraint. The government finances defense spending with proportional taxes on labor and capital income. I abstract from government debt in order to focus on the choice between labor and capital taxation and, in particular, on how the taxation of capital interacts with the government's incentive to protect that capital.

The government balances its budget in every period:

$$d_t = \tau_t w_t n_t + \theta_t r_t K_t. \tag{11}$$

Resources. Defense uses the same final good as private consumption and investment. The aggregate resource constraint is

$$c_t + x_t + d_t = F(K_t, n_t). \tag{12}$$

3.2 Timing

The timing of events and the interaction between the government and private agents within each period are as follows.

1. The economy enters period t with inherited capital K_t. The current geopolitical threat state z_t, and hence the war probability φ_t, is observed.

2. Current fiscal policy (τ_t, θ_t, d_t) is implemented. Under commitment, this policy is predetermined by the state-contingent plan chosen at date 0; under discretion, the government chooses current policy after observing the state.

3. Taking current government policy and factor prices as given, the household chooses consumption c_t, labor supply n_t, and productive investment x_t.

4. Production takes place, taxes are collected and defense expenditure d_t is implemented.

5. The destruction state $s_t \in \{P, W\}$ is realized. If $s_t = P$, total exposed capital survives. If $s_t = W$, only the fraction $S(d_t)$ of exposed capital survives.

6. The innovation to geopolitical risk is realized, determining z_{t+1} and therefore φ_{t+1}. The economy enters the next period with (K_{t+1}, z_{t+1}).

3.3 Household Problem

Let h^t denote the history of realized destruction states and geopolitical risk innovations through period t. Conditional on the initial state (K_0, φ_0), the history determines the current state (K_t, φ_t).

The household takes as given the history-contingent sequences

$$\left\{\tau_t(h^t), \theta_t(h^t), d_t(h^t), w_t(h^t), r_t(h^t)\right\}$$

and chooses

$$\left\{c_t(h^t), n_t(h^t), x_t(h^t)\right\}_{t\geq 0, h^t}$$

to maximize

$$\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t u\big(c_t(h^t), n_t(h^t)\big)\right]. \tag{13}$$

For $t \geq 1$, the household budget constraint is

$$c_t(h^t) + x_t(h^t) = \left[1-\tau_t(h^t)\right] w_t(h^t) n_t(h^t) + \left[1-\theta_t(h^t)\right] r_t(h^t) K_t(h^t). \tag{14}$$

At $t = 0$, I impose that the initial capital stock can not be taxed [1], so that

$$c_0 + x_0 = (1-\tau_0) w_0 n_0 + r_0 K_0. \tag{15}$$

Capital evolves according to

$$K_{t+1}(h^{t+1}) = \begin{cases} (1-\delta)K_t(h^t) + x_t(h^t), & \text{if } s_t = P, \\ S\big(d_t(h^t)\big)\left[(1-\delta)K_t(h^t) + x_t(h^t)\right], & \text{if } s_t = W. \end{cases} \tag{16}$$

The intratemporal labor condition is

$$-u_{n,t} = u_{c,t}(1-\tau_t) w_t. \tag{17}$$

The Euler equation for productive investment is

$$u_{c,t} = \beta \sum_{j\in\{H,L\}} \pi_j \Big\{ (1-\varphi_t) u^{P,j}_{c,t+1}\left[1-\delta+(1-\theta^{P,j}_{t+1}) r^{P,j}_{t+1}\right] + \varphi_t S(d_t) u^{W,j}_{c,t+1}\left[1-\delta+(1-\theta^{W,j}_{t+1}) r^{W,j}_{t+1}\right] \Big\}. \tag{18}$$

[1] Following the Ramsey taxation literature, I rule out taxation of initial wealth. Allowing it would induce government to tax that wealth as it would effectively be a non-distortionary lump sum tax

3.4 Competitive Equilibrium

Given an initial state (K_0, φ_0), a stochastic process for geopolitical risk, and a history-contingent government policy

$$\left\{\tau_t(h^t), \theta_t(h^t), d_t(h^t)\right\}_{t\geq 0, h^t},$$

a competitive equilibrium is a collection of history-contingent allocations

$$\left\{c_t(h^t), n_t(h^t), x_t(h^t), K_{t+1}(h^{t+1})\right\},$$

and factor prices

$$\left\{w_t(h^t), r_t(h^t)\right\},$$

such that:

1. Given prices and government policy, the household solves its optimization problem;
2. Firms maximize profits, so that

$$w_t = F_n(K_t, n_t), \qquad r_t = F_K(K_t, n_t);$$

3. The government budget constraint holds,

$$d_t = \tau_t w_t n_t + \theta_t r_t K_t; \tag{19}$$

4. The aggregate resource constraint holds,

$$c_t + x_t + d_t = F(K_t, n_t);$$

5. Capital evolves according to (16).

4 A Deterministic Economy

The stochastic economy described above provides the main environment of the paper. To make the mechanism transparent, I first consider a special case in which the destruction state occurs with certainty, $\varphi_t = 1$ for all t. This special case preserves the interaction between productive investment, defense expenditure, and taxation while removing uncertainty over the realization of destruction. I use this special case to characterize the main mechanism and comparative statics before returning to the stochastic economy in the quantitative analysis in Section 6.

In this section, I assume that the economy is initially in a no threat steady state, ie, the destruction probability $\varphi = 0$. At date $t = 0$, it experiences an unexpected permanent increase in destruction risk such that

$$\varphi_t = 1, \qquad t \geq 0.$$

The initial capital stock K_0 is inherited from no-threat steady state. The destruction state from date 0 onward is realized with certainty each period, so that the law of motion for capital becomes

$$K_{t+1} = S(d_t)\left[(1-\delta)K_t + x_t\right].$$

This special case essentially removes uncertainty over the realization of destruction while preserving the key interaction between productive investment, protective expenditure, and taxation. It provides a straightforward environment for characterizing the mechanism and comparative statics.[2]

4.1 Timing

The sequence of events order is the same the stochastic case, except that there is no innovation on z_t, therefore φ_t. Hence, destruction $S(d_t)$ will occur for sure. Since the paper will consider both the Ramsey case and the discretion case, there are two types of timing, and the only difference comes from when government policy plans are made. Commitment fixes the entire policy sequence at $t = 0$, while under discretion, policy is re-optimized after capital has been installed. The figure 4 illustrates the timing under commitment and discretion.

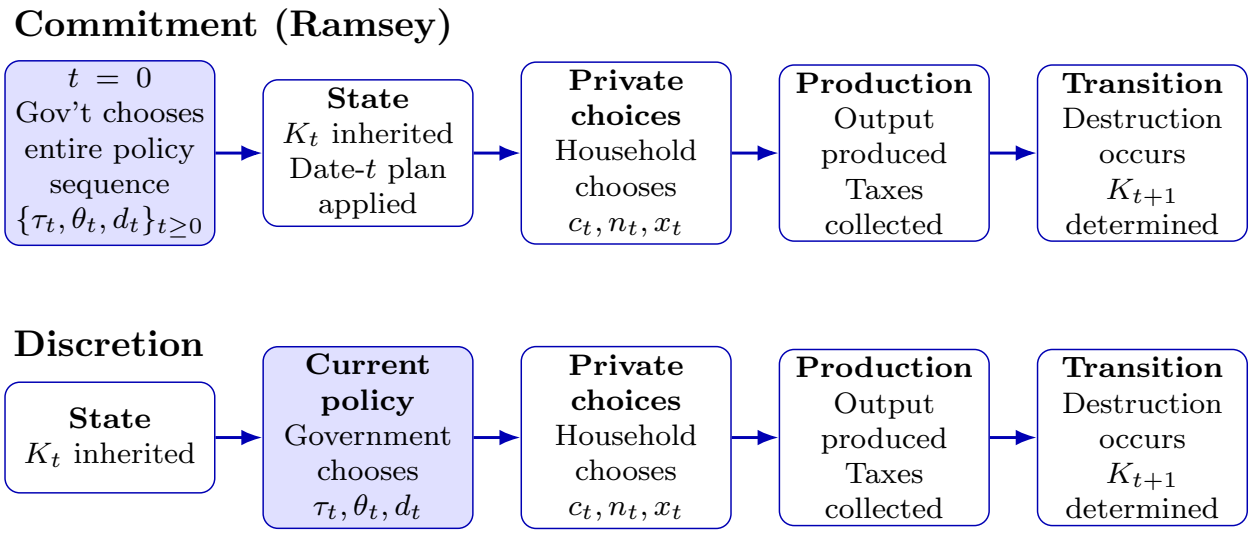


Figure 4: Timing under commitment and discretion

4.2 First Best and Intuition

I first characterize the efficient allocation. This benchmark isolates the basic tradeoff between productive investment x_t and defense expenditure d_t. The two uses of resources compete contemporaneously, but are dynamically complementary. Productive investment increases the stock of

[2]The mechanism extends beyond deterministic destruction. More generally, $S(d)$ can be interpreted as expected survival, for example $S(d) = 1 - p(d)\ell(d)$, where $p(d)$ is the probability of conflict and $\ell(d)$ is destruction conditional on conflict. Defense may increase survival through deterrence, mitigation, or both.

capital available for future production, while defense increases the fraction of both existing and newly accumulated capital that survives destruction.

Given K_0, a benevolent planner chooses $\{c_t, n_t, x_t, d_t\}_{t=0}^{\infty}$ to solve

$$\max_{\{c_t,n_t,x_t,d_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, n_t) \tag{20}$$

subject to, for every t,

$$c_t + x_t + d_t = F(K_t, n_t), \tag{21}$$

$$K_{t+1} = S(d_t)\left[(1-\delta)K_t + x_t\right]. \tag{22}$$

For an interior solution, the first-order conditions for productive investment and defense are

$$u_{c,t} = \beta u_{c,t+1}\left[1 + F_{K,t+1} - \delta\right] S(d_t), \tag{23}$$

$$u_{c,t} = \beta u_{c,t+1}\left[1 + F_{K,t+1} - \delta\right] S'(d_t)\left[(1-\delta)K_t + x_t\right]. \tag{24}$$

Because both choices have the same marginal cost and share the same continuation value of capital, combining (23) and (24) gives

$$S(d_t^*) = \left[(1-\delta)K_t + x_t^*\right] S'(d_t^*). \tag{25}$$

In other words, it is optimal to invest in x and d up to the point where their marginal return are equalized at each date. Note that Equation (25) alone does not determine the optimal pair (x_t^*, d_t^*); it characterizes one optimality relation between the two controls.

Lemma 1 (Complementarity between Productive Investment and Defense)**.** *Suppose the first-best allocation is interior and*

$$S'(d) > 0, \qquad S''(d) \leq 0.$$

For a given inherited capital stock K, the first-best optimality locus between productive investment and defense is upward sloping:

$$\frac{\partial x^*(d;K)}{\partial d} = 1 - \frac{S(d)S''(d)}{[S'(d)]^2} \geq 1 > 0. \tag{26}$$

Hence, conditional on inherited capital, greater defense is associated with greater productive investment along the first-best optimality locus.

Greater defense raises the survival rate of newly accumulated capital and therefore increases the effective return to productive investment. Conversely, greater productive investment expands the amount of capital exposed to destruction and raises the marginal return to defense. In this sense, productive and protective investment are complementary along the first-best allocation.

Under functional form

$$S(d) = 1 - ae^{-bd},$$

the interior condition is

$$d^*(x; K) = \frac{1}{b} \log \left[a \left(1 + b\big((1-\delta)K + x\big) \right) \right], \tag{27}$$

with

$$\frac{\partial d^*(x; K)}{\partial x} = \frac{1}{1 + b\big((1-\delta)K + x\big)} > 0. \tag{28}$$

Figure 5 shows that for a given inherited capital stock K, greater productive investment increases the amount of capital exposed to destruction and raises optimal defense expenditure.

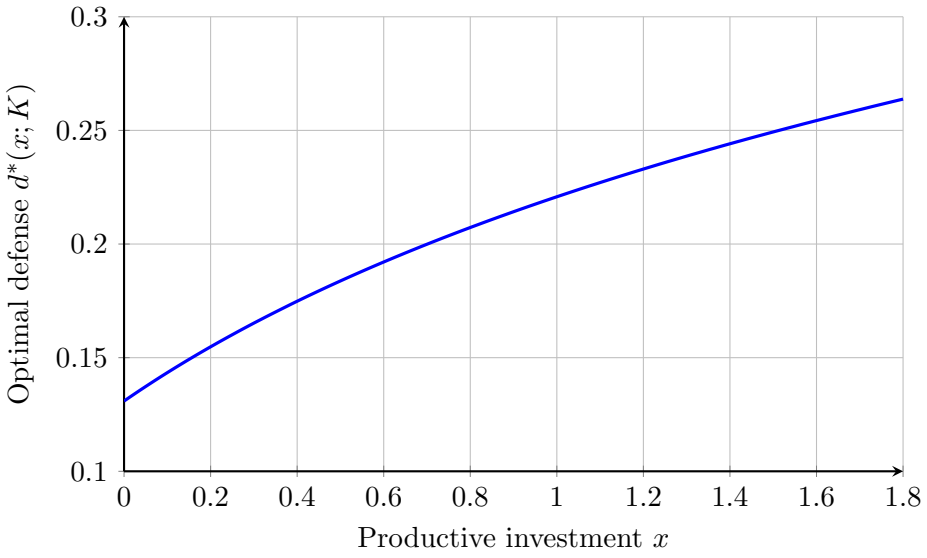


Figure 5: First-best relationship between productive investment and defense

Proposition 1 (Threshold for positive defense)**.** *Suppose* $S(d) = 1 - ae^{-bd}$ *with* $a \in (0, 1)$ *and* $b > 0$*. For given* a *and* b*, there exists a threshold level of exposed capital* $A^*(a, b) = (1 - a)/(ab)$ *such that* $d^* = 0$ *for* $A \leq A^*(a, b)$ *and* $d^* > 0$ *for* $A > A^*(a, b)$*.*

Proof. See Appendix A.1.

Equivalently, positive defense is optimal if and only if $a(1 + bA) > 1$. The threshold $A^*(a, b)$ is decreasing in both destruction exposure a and defense effectiveness b. Moreover,

$$\frac{\partial A^*}{\partial a} = -\frac{1}{a^2 b} < 0, \qquad \frac{\partial A^*}{\partial b} = -\frac{1-a}{ab^2} < 0.$$

These equations mean: when destruction exposure a is higher, the threshold level of exposed capital required to justify positive defense is lower; when defense is more effective, the threshold is also lower.

Proposition 1 shows that defense provision depends jointly on the amount of capital exposed

to destruction, the severity of destruction, and the effectiveness of defense. When exposed capital is small, destruction is not severe or defense is not sufficiently effective, the marginal gain from the first unit of defense may not justify its resource cost.

Figure 6 illustrates the threshold mechanism. The marginal physical return to productive investment is $S(d)$, whereas the marginal physical return to defense is $AS'(d)$. Positive defense is optimal only when the marginal return to defense is sufficiently high relative to the return to productive investment. A larger stock of exposed capital shifts $AS'(d)$ upward, making positive defense more likely.

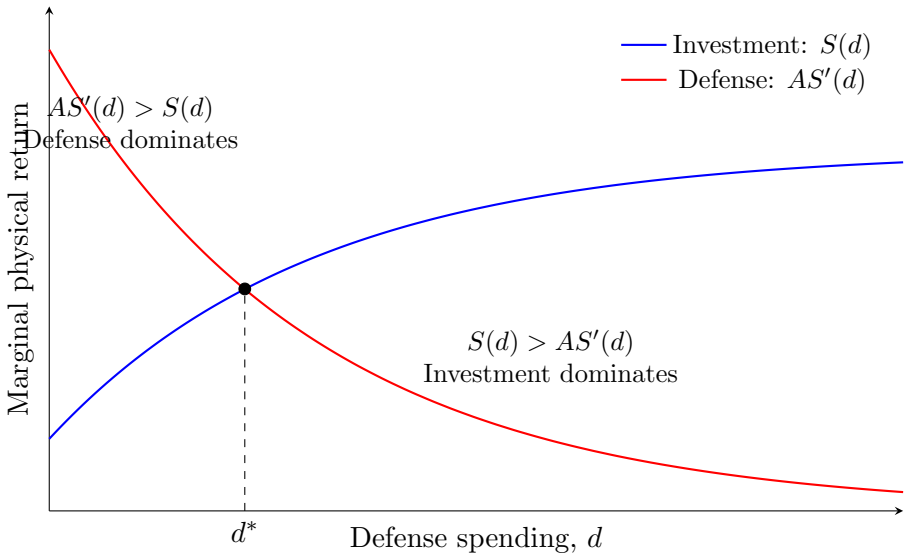


Figure 6: Marginal physical returns to investment and defense. An additional unit of investment increases next-period capital by $S(d)$, while an additional unit of defense increases next-period capital by $AS'(d)$, where $A = (1-\delta)K + x$ is the exposed capital. The efficient interior allocation d^* satisfies $S(d^*) = AS'(d^*)$.

4.2.1 Steady-State Adaptation Criterion

Proposition 1 characterizes when defense becomes positive. I now characterize when the endogenous defense response is sufficiently strong to reverse the decline in productive capacity caused by greater destruction exposure. For this purpose, I specialize the production technology to

$$F(K, n) = K^{\alpha} n^{1-\alpha}, \qquad \alpha \in (0, 1),$$

and retain the exponential survival technology

$$S(d, a) = 1 - ae^{-bd}, \qquad a \in (0, 1), \quad b > 0.$$

The result does not require a particular functional form for period utility beyond the assumptions imposed above.

Proposition 2 (Adaptation criterion)**.** *Suppose production is Cobb–Douglas, $\beta \in (0,1)$, $\delta \in [0,1]$, and the survival technology is*

$$S(d,a) = 1 - ae^{-bd}.$$

Consider a differentiable, interior first-best steady state, and define

$$d_a^{ss} \equiv \frac{d\, d^{ss}(a)}{da}.$$

Effective survival satisfies

$$\frac{dS^{ss}}{da} = e^{-bd^{ss}} \left(abd_a^{ss} - 1\right).$$

Moreover, steady-state capital, productive investment, and output are strictly increasing functions of effective survival.

Consequently,

$$\frac{dK^{ss}}{da} > 0, \qquad \frac{dx^{ss}}{da} > 0, \qquad \frac{dY^{ss}}{da} > 0$$

if and only if

$$abd_a^{ss} > 1.$$

If $abd_a^{ss} < 1$, all three instead decline with destruction exposure.

Proof. See Appendix A.2. □

To interpret this condition, define the adaptation statistic

$$\mathcal{A}(a) \equiv ab\, d_a^{ss}.$$

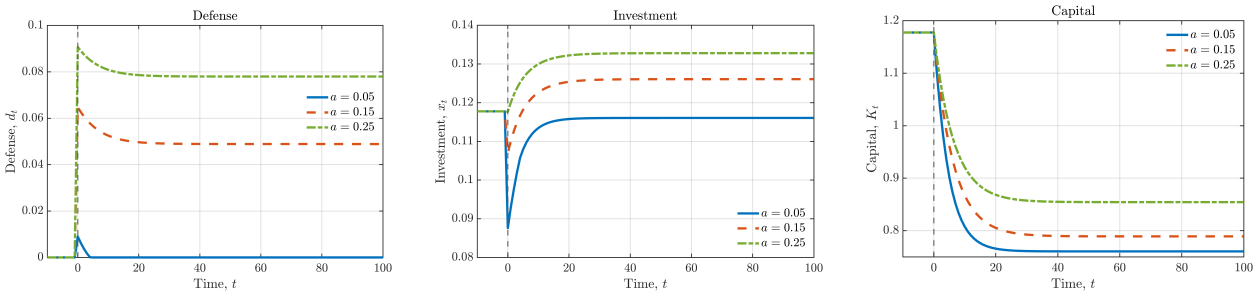
This statistic measures whether the endogenous defense response is sufficient to offset the direct effect of greater destruction exposure on effective survival. If $\mathcal{A}(a) < 1$, protection responds insufficiently: effective survival falls, and steady-state productive investment, capital, and output decline. If $\mathcal{A}(a) = 1$, defense exactly offsets the marginal increase in destruction exposure, leaving effective survival and productive capacity unchanged. Finally, if $\mathcal{A}(a) > 1$, defense more than offsets the increase in exposure. Effective survival rises, and productive investment, capital, and output recover.

4.2.2 Numerical Illustration

Lemma 1 characterizes the local relationship between productive investment and defense. I next show numerically how complementarity affects the economy's transition and long-run response to destruction. For the survival technology, I use

$$a = 0.15, \qquad b = 20,$$

as the benchmark values. A detailed discussion of the parameters and calibration of a and b is provided in Section 5.1. I vary these parameters in the comparative statics below.



(a) Defense (b) Investment (c) Capital

Figure 7: First-best transition dynamics

Figure 7 shows the response of the economy to a permanent and certain destruction. The economy enters the new environment with a relatively large inherited capital stock. Because more capital is immediately exposed to destruction, the marginal return to protection is high and the planner front-loads defense expenditure. As capital adjusts to the new environment, the amount of exposed capital falls and defense expenditure declines from its initial peak. The blue line in the first panel is consistent with Proposition 1: defense may not be worthwhile when destruction exposure is sufficiently low. Investment initially falls because investment and defense compete for current resources. As higher defense raises the post-destruction return to investment, productive investment recovers. The complementarity between productive and protective investment is stronger when destruction exposure is greater, leading to larger responses of both defense and investment.

The same mechanism appears in the long-run comparative statics. Figure 8 plots steady-state defense and investment at varying destruction exposure. At low levels of a, optimal defense remains at the corner $d^{ss} = 0$. Once destruction becomes sufficiently severe to make protection worthwhile, defense rises and the complementarity between protection and accumulation induces greater productive investment. The steady-state response also depends strongly on defense effectiveness. Holding destruction exposure fixed, lower defense effectiveness (blue dashed line) raises the threshold for positive defense and therefore expands the zero-defense region. This pattern is consistent with Proposition 1.

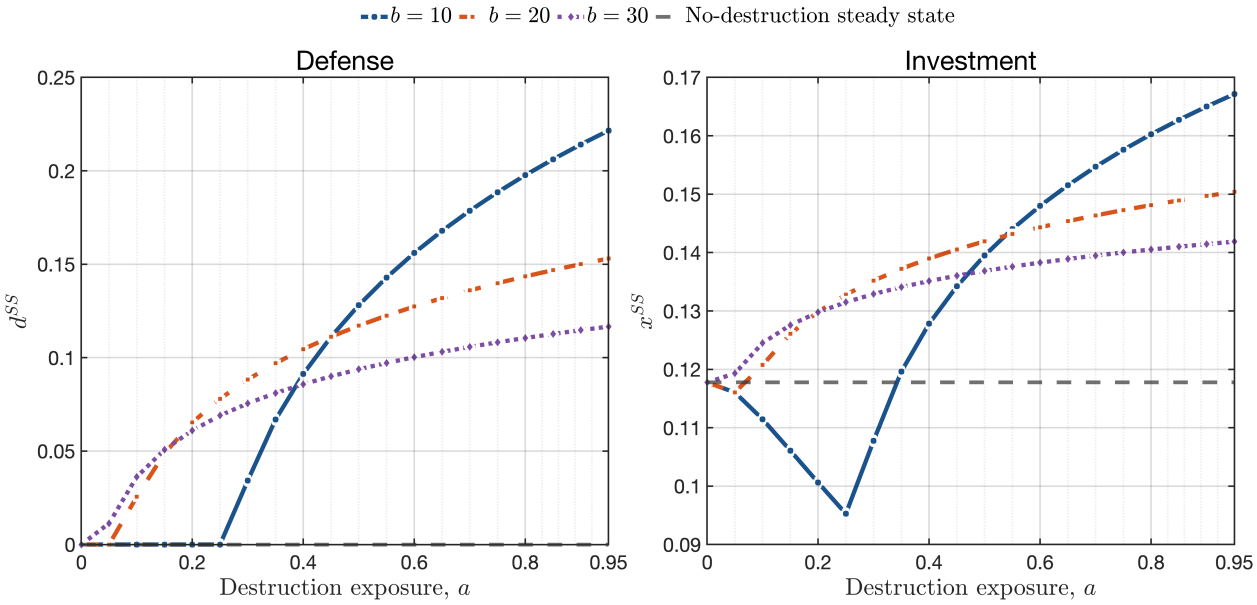


Figure 8: Steady-state defense and investment

Figure 9 shows the corresponding response of steady-state capital and output. When destruction exposure is low and defense is not provided, greater destruction reduces productive capacity. As destruction exposure rises, optimal defense and productive investment increase. When defense is sufficiently effective, these endogenous responses can partially offset the decline in steady-state capital and output.

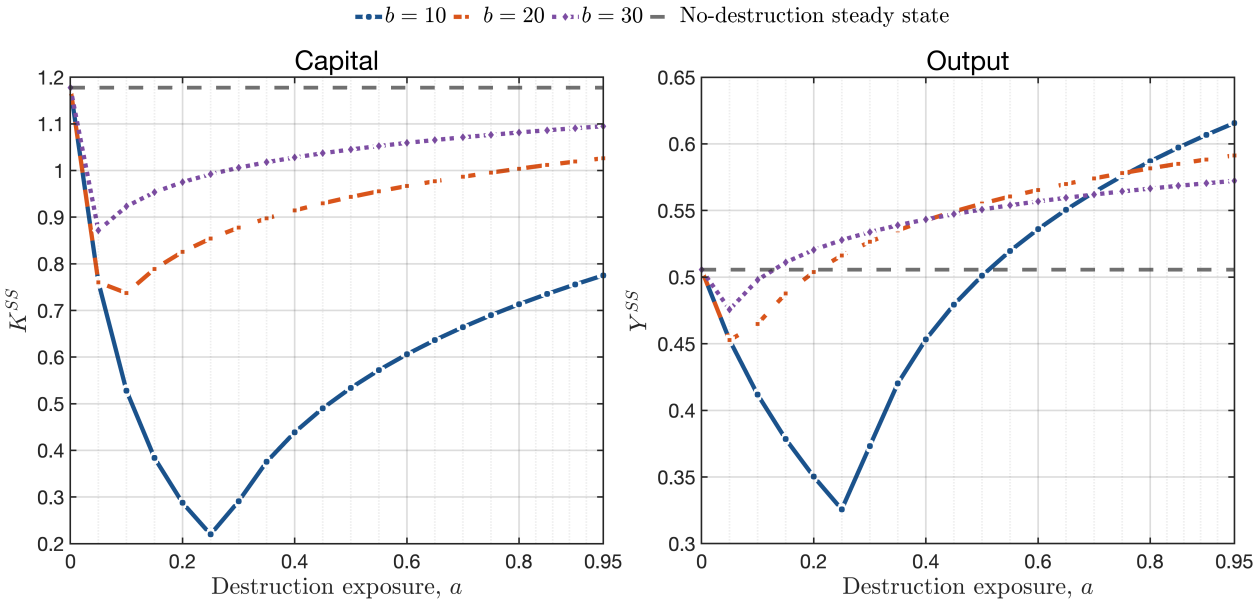


Figure 9: Steady-state capital and output

Result 1 (Adaptation)**.** *For the benchmark parameterization, steady-state capital and output are nonmonotonic in destruction exposure. At low levels of destruction exposure, greater destruction*

reduces capital and output. Once protection becomes sufficiently useful, however, optimal defense and productive investment increase, partially reversing the decline in productive capacity.

This adaptation is costly. Sustaining productive capacity requires greater defense expenditure and higher labor supply, while consumption remains restrained. Welfare continues to decline even as capital and output recover.

Adaptation requires a substantial reallocation of resources. Geopolitical risk acts like a costly constraint on the economy's preferred allocation. Figure 10 shows that the efficient response is to sacrifice leisure and consumption and redirect resources toward production, investment, and protection. Absent threat, household would have preferred to consume more and work less. When defense is less effective, this adjustment is more costly: households work substantially more, consume less, and a larger share of output must be devoted to defense. In other words, maintaining a high level of productive capacity requires a larger sacrifice of consumption and leisure when protection is less effective.

Figure 11 confirms that welfare declines with destruction exposure, including over regions in which capital and output recover. For a given level of destruction exposure, less effective defense requires greater labor supply and larger allocations to investment and protection, leaving less for consumption. Welfare is therefore lower when the defense technology is less effective.

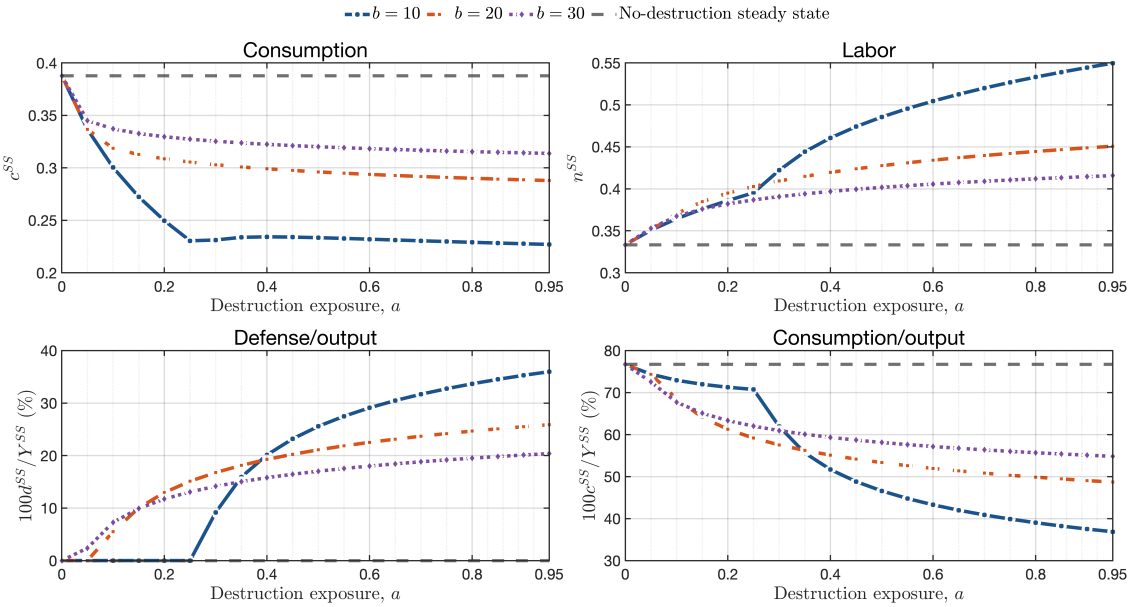


Figure 10: Steady-state consumption, labor, defense expenditure, and consumption shares

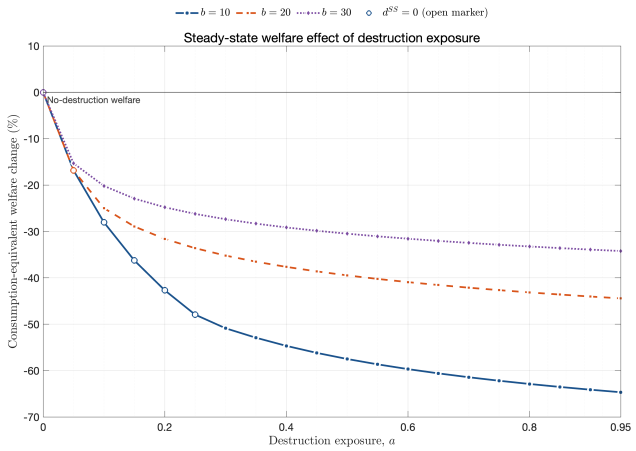


Figure 11: Welfare loss increases with destruction exposure

Notes: The figure reports first-best steady-state welfare as destruction exposure a varies for selected values of defense effectiveness b. Welfare declines even in regions where endogenous protection allows capital and output to recover.

The strength of this endogenous adaptation depends on the specification of the defense technology. In the baseline specification $S(d)$, a larger capital stock does not require proportionally greater defense expenditure to maintain a given survival rate. Appendix B considers an alternative specification $S(d/K)$. When protection depends on defense expenditure relative to the capital stock, expansion of K dilutes protection for a given level of d. That weakens or eliminates the non-monotonic result of steady-state capital and output.

4.3 Ramsey Government

I now ask how much of this adaptation remains feasible when defense must be financed through distortionary labor and capital taxation. I first consider a government that can commit at date 0 to a complete sequence of future fiscal policies. The Ramsey government chooses the fiscal plan whose induced competitive equilibrium maximizes household welfare.

Let

$$\pi = \{\tau_t, \theta_t, d_t\}_{t=0}^{\infty}$$

denote a fiscal plan. Given K_0, the Ramsey government solves

$$\max_{\pi} \sum_{t=0}^{\infty} \beta^t u(c_t, n_t), \tag{29}$$

where the allocation induced by π must constitute a competitive equilibrium.

Unlike first-best, the Ramsey government chooses taxes and defense, while consumption, labor, and productive investment are private equilibrium responses. The government therefore internalizes how current and anticipated future taxation affects household saving and the accumulation of the capital that defense is intended to protect.

4.3.1 Competitive Equilibrium and Implementability

Under the deterministic environment considered in this section, household optimality requires

$$-u_{n,t} = u_{c,t}(1-\tau_t)w_t, \tag{30}$$

$$u_{c,t} = \beta u_{c,t+1} S(d_t)\left[(1-\delta) + (1-\theta_{t+1})r_{t+1}\right]. \tag{31}$$

Competitive firms imply

$$w_t = F_n(K_t, n_t), \qquad r_t = F_K(K_t, n_t). \tag{32}$$

The government budget constraint is

$$d_t = \tau_t w_t n_t + \theta_t r_t K_t, \tag{33}$$

and market clearing implies

$$c_t + x_t + d_t = F(K_t, n_t). \tag{34}$$

Capital evolves according to

$$K_{t+1} = S(d_t)\left[(1-\delta)K_t + x_t\right]. \tag{35}$$

As is standard in the Ramsey problem, I rule out taxation of the initial capital stock by imposing $\theta_0 = 0$.

Combining the household budget constraint with the household optimality conditions eliminates taxes and prices and yields the implementability constraint. For $t \geq 1$,

$$\beta u_{c,t}\left[c_t + x_t + (1-\delta)K_t\right] + \beta u_{n,t} n_t = u_{c,t-1}\left[(1-\delta)K_{t-1} + x_{t-1}\right]. \tag{36}$$

The derivation is provided in Appendix C.0.2.

The implementability constraint captures the intertemporal restriction imposed by private saving behavior. The allocation chosen at date t cannot be selected independently of past decisions: the right-hand side of (36) is inherited from the previous period.

Here, I follow the recursive approach to dynamic optimal taxation in Kydland and Prescott (1980). Define the inherited implementability state

$$q_t \equiv u_{c,t-1}\left[(1-\delta)K_{t-1} + x_{t-1}\right]. \tag{37}$$

The implementability constraint becomes

$$\beta u_c(c_t, n_t)\left[c_t + x_t + (1-\delta)K_t\right] + \beta u_n(c_t, n_t) n_t = q_t, \tag{38}$$

while current saving determines next period's implementability state,

$$q_{t+1} = u_c(c_t, n_t)\left[(1-\delta)K_t + x_t\right]. \tag{39}$$

Thus, in addition to the physical capital stock K_t, the Ramsey planner must keep track of the inherited implementability state q_t. This state summarizes the restriction imposed by past household saving decisions on the allocation that can be implemented today. Household optimality therefore allows taxes and prices to be eliminated from the government's problem. The Ramsey allocation can equivalently be characterized as the solution to a primal problem in which the government chooses allocations directly subject to feasibility, capital accumulation, and implementability.

4.3.2 Recursive Ramsey Problem

The recursive state is (K, q). The Ramsey problem can be written as

$$V^R(K, q) = \max_{c,n,x,d} \left\{ u(c, n) + \beta V^R(K', q') \right\} \tag{40}$$

subject to

$$c + x + d = F(K, n), \tag{41}$$

$$K' = S(d) \left[(1 - \delta)K + x \right], \tag{42}$$

$$\beta u_c(c, n) \left[c + x + (1 - \delta)K \right] + \beta u_n(c, n)n = q, \tag{43}$$

$$q' = u_c(c, n) \left[(1 - \delta)K + x \right]. \tag{44}$$

As in the first-best problem, define

$$A \equiv (1 - \delta)K + x.$$

The two endogenous state transitions can then be written as

$$K' = S(d)A, \qquad q' = u_c(c, n)A. \tag{45}$$

This representation makes the difference from the first best clear. Investment affects both next-period productive capital K' and next-period implementability q'. Defense instead reduces expected destruction and therefore affects the amount of physical capital that survives. The Ramsey government must take these fiscal implementation effects into account when allocating resources between investment and protection. The treatment of the initial period, the full derivation of the implementability constraint, tax recovery, and numerical solution are provided in Appendix C.

4.4 Policy with No Commitment

I next consider a government that cannot commit to future fiscal policy. In each period, the government inherits the current capital stock K and chooses current labor and capital income taxes and defense expenditure, taking as given that future governments will reoptimize. Because past investment is already embodied in K, the current government does not internalize the effect of its current capital tax on the saving decision that generated the inherited capital stock.

I characterize the discretionary allocation as a stationary Markov-perfect equilibrium. Let

$$\Gamma(K) = \{\tau(K), \theta(K), d(K), c(K), n(K), x(K)\}$$

denote stationary policy and allocation rules as functions of the current capital stock. Given continuation rules $\Gamma(\cdot)$, the government at state K solves

$$V^D(K) = \max_{\tau,\theta,d,c,n,x} \left\{u(c,n) + \beta V^D(K')\right\}, \tag{46}$$

subject to

$$c + x + d = F(K,n), \tag{47}$$

$$K' = S(d)\left[(1-\delta)K + x\right], \tag{48}$$

$$d = \tau F_n(K,n)n + \theta F_K(K,n)K, \tag{49}$$

together with household optimality,

$$-u_n(c,n) = u_c(c,n)(1-\tau)F_n(K,n), \tag{50}$$

$$u_c(c,n) = \beta u_c(c',n')S(d)\left[1 - \delta + (1-\theta')F_K(K',n')\right], \tag{51}$$

where

$$(c', n', \theta') = \big(c(K'),\, n(K'),\, \theta(K')\big)$$

are determined by the continuation equilibrium.

A stationary Markov-perfect equilibrium is a set of policy and allocation rules $\Gamma(K)$ such that, at every state K, the government's current choices solve the problem above given the same continuation rules $\Gamma(\cdot)$. Thus, unlike the Ramsey government, the discretionary government takes future policy rules as given rather than choosing the entire future policy sequence at date zero.

This distinction is the source of the standard time-consistency problem in capital taxation. Current capital K is predetermined when the government chooses θ, whereas households use the continuation rule $\theta(K')$ when choosing x. The current government therefore has an incentive to tax installed capital, while households anticipate that future governments will have the same incentive.

4.5 Mechanism

Before I continue to the quantitative results in the next section and analyze the allocations and tax structure under the two regimes, it is useful to illustrate how capital taxation may amplify the welfare loss.

In the standard capital-tax problem, expected future capital taxes reduces household saving and productive investment, which leads to a smaller capital stock. In the present model, this effect is amplified by a protection channel. A lower capital stock reduces the value of defense, because there is less productive capital to protect. As a result, the government optimally provides less defense,

which lowers the survival rate of capital in adverse events. The higher expected destruction further reduces the effective return to investment, discouraging capital accumulation even more. This feedback loop therefore strengthens the initial distortion from capital taxation and generates an additional cost relative to a setting absent of protection.

The figure below illustrates this mechanism. The standard channel operates through the box on the left. The protection channel is highlighted on the right.

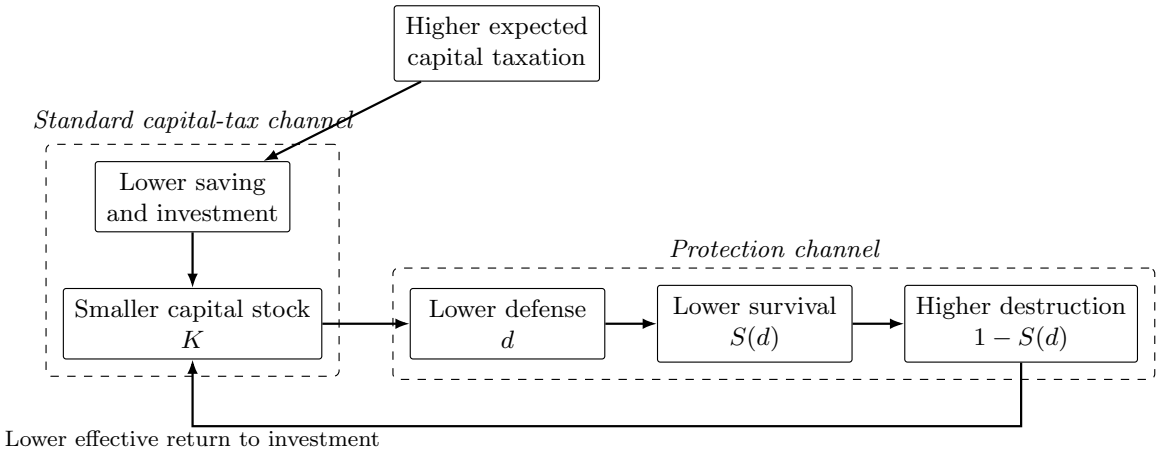


Figure 12: Capital taxation and the protection channel.

5 Quantitative Results

I now use the quantitative model to compare the economy's response to destruction exposure under the first best, Ramsey commitment, and discretion. The analysis focuses on two dimensions. First, I compare transition dynamics following an increase in destruction exposure. Second, I compare the long-run allocations across the three policy regimes.

The first-best allocation provides the efficient benchmark. Comparing Ramsey with the first best isolates the cost of financing defense through distortionary taxation, while comparing Ramsey with discretion isolates the additional cost of limited commitment.

5.1 Calibration

The model is annual. I set the discount factor to $\beta = 0.96$, the capital share to $\alpha = 0.33$, and the depreciation rate to $\delta = 0.10$. Preferences are given by

$$u(c, n) = \frac{c^{1-\sigma} - 1}{1 - \sigma} - \chi \frac{n^{1+1/\phi}}{1 + 1/\phi}.$$

I set the coefficient of relative risk aversion to $\sigma = 1.5$ and the Frisch elasticity parameter to $\phi = 1$. The labor-disutility weight χ is choosen so that steady-state labor supply is $\bar{n} = \frac{1}{3}$.

Calibration of the Destruction Technology a and b. Define the destruction rate conditional on a destructive event as

$$\ell(d) \equiv 1 - S(d). \tag{52}$$

Under the exponential survival technology,

$$S(d) = 1 - ae^{-bd},$$

the destruction rate is

$$\ell(d) = ae^{-bd}. \tag{53}$$

The two parameters have distinct interpretations. Since

$$\ell(0) = a,$$

a measures destruction in the absence of protection, while b governs how rapidly destruction declines with defense expenditure.

Destruction severity. I discipline a using historical estimates of cumulative capital destruction during interstate conflicts. Table 1 summarizes estimates of cumulative capital destruction in selected interstate conflicts. Across these episodes, realized losses are generally between about 5 and 25 percent of the pre-war capital stock, with substantial heterogeneity across conflicts. These observations do not directly identify a, because realized destruction occurs in economies that devote positive resources to defense. I therefore use the historical evidence to discipline the plausible magnitude of the zero-protection destruction parameter rather than to estimate it directly. I consider

$$a \in [0.05, 0.25]$$

and set

$$a = 0.15$$

in the benchmark calibration. The benchmark lies near the middle of the historical range. The lower end of the range represents relatively limited capital destruction and the upper end is motivated by severe historical episodes such as Japan during World War II, for which estimates of aggregate capital destruction are close to one quarter of the pre-war capital stock.

Table 1: Capital Destruction and Military Spending in Selected Conflicts

Country / conflict	War year	Capital destruction	Military spending / GDP
Finland	1939	7.4%	1.50%
Germany, WWII	1939	17.1%	8.30%
France, WWII	1940	23.0%	3.90%
Netherlands	1940	7.0%	3.97%
Norway	1940	8.1%	1.17%
U.K.	1940	5.0%	3.50%
Japan*	1944	25.8%	24.40%
Ukraine	2022	10.0%	3.97%

Notes: Capital destruction is cumulative and is measured relative to the pre-war capital stock. Data are from Luetticke et al. (2025). Military expenditure is measured as a share of GDP using the Global Military Spending Dataset of Barnum et al. (2025). For all observations other than Japan, military spending is averaged over the three years preceding the conflict. For Ukraine, the pre-war average corresponds to 2019–2021. *For Japan, the reported military-spending average covers 1941–1943 and therefore reflects wartime rather than strictly pre-war expenditure. The Japan observation is shown only as an upper-end historical reference.

Defense effectiveness. I discipline the effectiveness parameter b by matching a target destruction rate at a representative level of defense expenditure. Given the benchmark value of a, equation (53) implies

$$b = -\frac{\log\left(\ell^{\text{target}}/a\right)}{d^{\text{target}}}. \tag{54}$$

I choose ℓ^{target} and d^{target} to represent the annual destruction rate and defense expenditure during a destructive geopolitical episode. Because observed military spending responds endogenously to the severity of conflict, this exercise should be interpreted as a calibration target rather than a causal estimate of defense effectiveness.

Ukraine provides a useful benchmark. The capital destruction during the first year of the full-scale conflict was approximately 10 percent of the pre-war capital stock. Because military capacity cannot adjust instantaneously after the onset of conflict, I use pre-war average military expenditure over 2019–2021 as a proxy for the effective defense resources available when the conflict began. This corresponds to military expenditure of approximately 3.97 percent of GDP. Conditional on the benchmark no-protection destruction rate $a = 0.15$, the implied value of b satisfies

$$0.10 = 0.15\exp[-b(0.0397)],$$

where output is normalized to one for this calculation. Hence,

$$b = -\frac{\log(0.10/0.15)}{0.0397} \simeq 10.2. \tag{55}$$

To assess whether this magnitude is reasonable, I repeat the same calculation for other historical episodes. Conditional on $a = 0.15$, each episode with realized destruction below the benchmark

no-protection destruction rate implies

$$b_i = -\frac{\log(\ell_i/a)}{d_i},$$

where d_i is pre-war military expenditure relative to GDP. The resulting values in table 2 exhibit substantial heterogeneity across episodes, ranging from about 10 to above 50. This variation is unsurprising: observed destruction reflects not only defense resources but also differences in attack intensity, destructive technology, geography, and other conflict-specific factors that are not separately modeled here.

Table 2: Defense Effectiveness Implied by Selected Historical Episodes

Country	Capital destruction	Military spending / GDP	Implied b_i
Finland	7.4%	1.50%	47.1
Netherlands	7.0%	3.97%	19.2
Norway	8.1%	1.17%	52.7
U.K.	5.0%	3.50%	31.4
Ukraine	10.0%	3.97%	10.2

Notes: Implied values are calculated from $b_i = -\log(\ell_i/a)/d_i$ using $a = 0.15$ and normalizing output to one. Pre-war military expenditure is used as a proxy for effective defense resources available at the onset of conflict. Episodes with realized destruction exceeding $a = 0.15$ are not used in this calculation, since they cannot be rationalized by variation in $b > 0$ conditional on a common $a = 0.15$. Such episodes may reflect greater underlying destruction intensity or more destructive attack technology. These implied values should be interpreted as calibration diagnostics rather than structural estimates of defense effectiveness.

Instead of using any single historical episode to identify b, I choose $b = 20$ as an intermediate benchmark within the range implied by the historical evidence and report comparative statics for alternative values of defense effectiveness. Economically, the benchmark $b = 20$ can be interpreted as an economy with a more effective protection technology than the Ukraine-implied lower-end case, but substantially below the effectiveness implied by several other historical episodes.

Table 3 summarizes the benchmark parameters.

Table 3: Benchmark Parameter Values

Parameter	Value	Description	Source / Target
β	0.96	Discount factor	Annual frequency
α	0.33	Capital share	Standard value
δ	0.10	Capital depreciation rate	Standard value
σ	1.50	Relative risk aversion	Standard value
ϕ	1.00	Frisch elasticity parameter	Standard value
χ	12.62	Labor disutility weight	$n^{SS} = 1/3$ when $a = 0$
a	0.15	Destruction exposure	Historical range $a \in [0.05, 0.25]$
b	20.00	Defense effectiveness	Calibrated to observed destruction rates

Notes: The labor-disutility parameter χ is calibrated so that labor supply equals one-third in the no-destruction steady state. Parameters a and b are varied in the comparative-static exercises.

5.2 Transition Dynamics Across Policy Regimes

I now compare the response to a permanent destruction environment, $\varphi_t = 1$ for all t, under the first best, Ramsey government, and discretionary government. The economy begins in the no-destruction steady state. At date $t = 0$, the economy enters a permanent destruction environment. The severity of destruction is governed by $a > 0$. I then vary a to study how the three regimes respond to different levels of destruction severity, holding all other parameters fixed.

Figure 13 compares the transition of defense, capital and output across the three regimes. All three regimes respond to the arrival of permanent destruction by increasing defense, but the magnitude of the response differs substantially. Defense is highest under the first best, lower under Ramsey, and lowest under discretion. Under all three regimes, the onset of permanent destruction generates an initial sharp increase in defense expenditure. This is because the economy enters the new environment with a relatively large inherited capital stock and the value of protecting existing productive capacity is high. Defense is therefore front-loaded. Productive investment initially falls as resources are reallocated toward protection, but subsequently recovers as higher defense raises the survival return to newly accumulated capital.

Capital declines in all three regimes following the shock, but the decline is smallest under the first best and largest under discretion. Output also has the same ranking. The Ramsey economy therefore preserves part of the first-best adaptation response, whereas the discretionary economy exhibits a substantially larger loss of productive capacity.

The differences in response across regimes widen during the transition and persist in the long run. Under discretion, the combination of weaker protection and lower capital accumulation is particularly harmful: the economy converges to substantially lower levels of defense, capital, and output than under the first best and Ramsey.

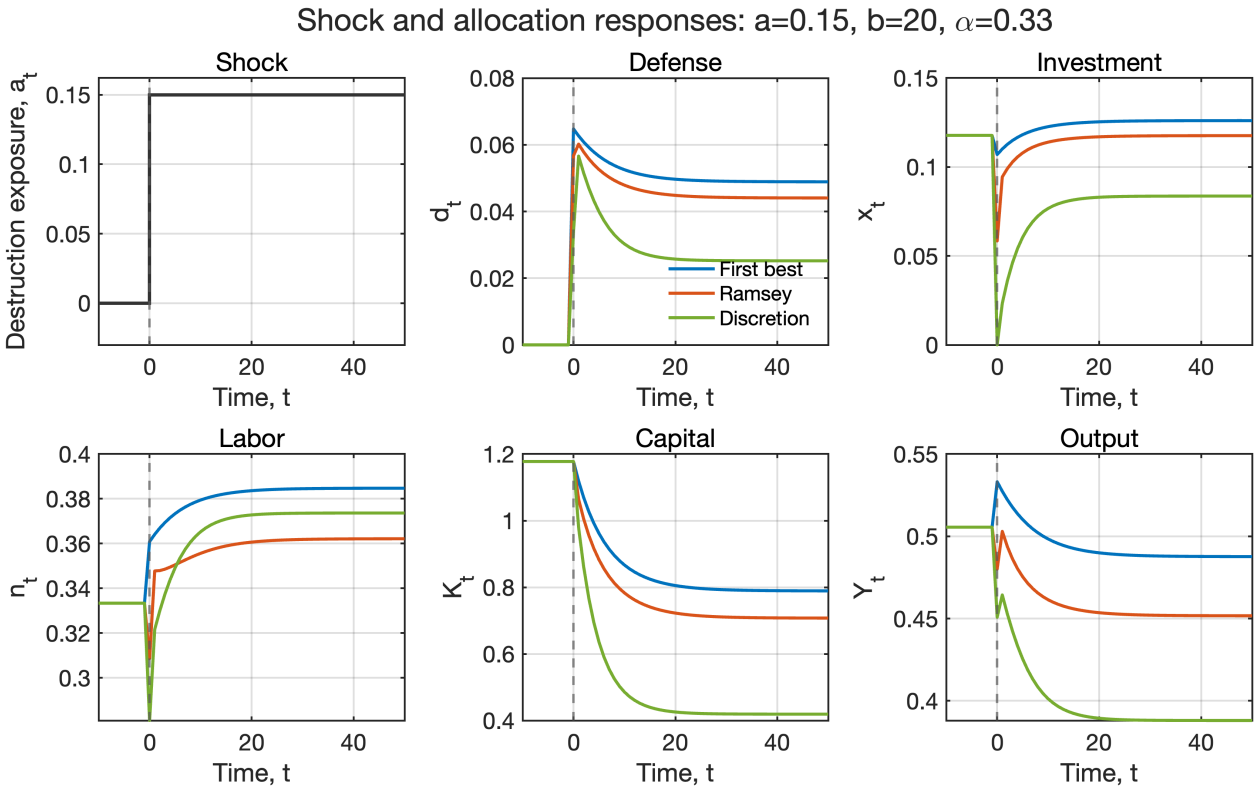


Figure 13: Transition dynamics under three regimes

The fiscal policies help explain the different transition paths under Ramsey policy and discretion. Figure 14 shows that Ramsey policy and discretion rely on very different tax structures during the rebuilding transition. The Ramsey government finances the increase in defense primarily through labor-income tax, while the discretionary government relies much more heavily on capital taxation. Moreover, during the rebuilding transition, when rebuilding productive capacity is especially valuable, the committed government chooses to subsidize capital accumulation, with a maximum subsidy of 0.29% and it peaks 16 years after the shock.

This difference in tax policy is central to the allocation dynamics. Under Ramsey policy, subsidizing capital and limiting capital taxation preserve the incentive to rebuild the productive base. Under discretion, taxation of installed capital generates the standard investment distortion: households anticipate lower after-tax returns and accumulate less capital. In this model, however, the effect is amplified by the protection channel: lower investment reduces the future capital stock, which in turn lowers the value of protection and weakens defense provision. Capital taxation therefore reduces productive capacity both directly through accumulation and indirectly through protection.

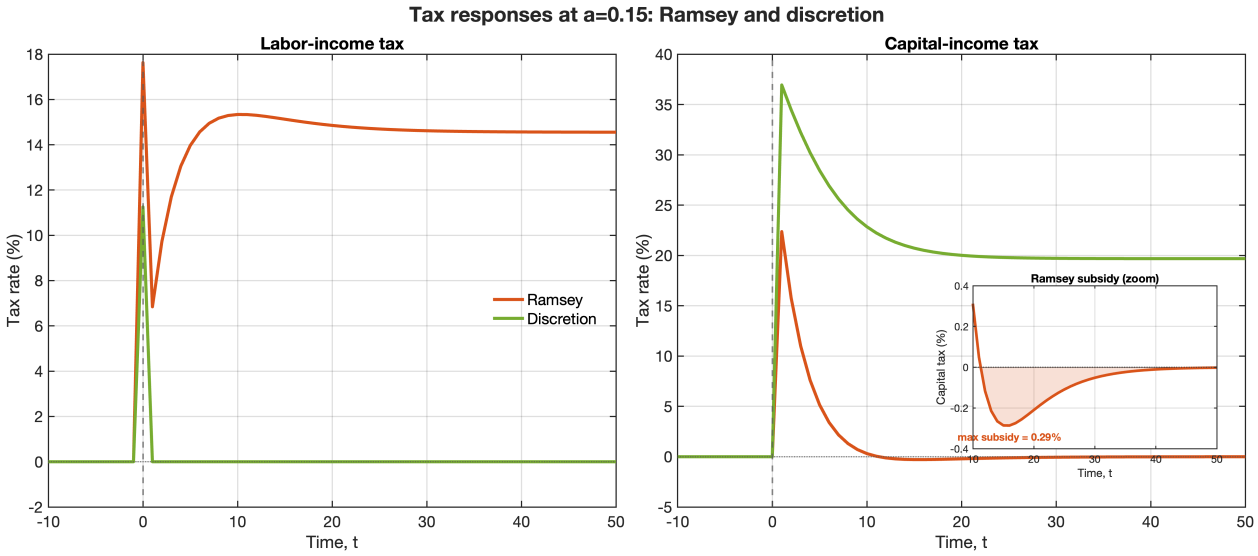


Figure 14: Tax transition under Ramsey and Discretion

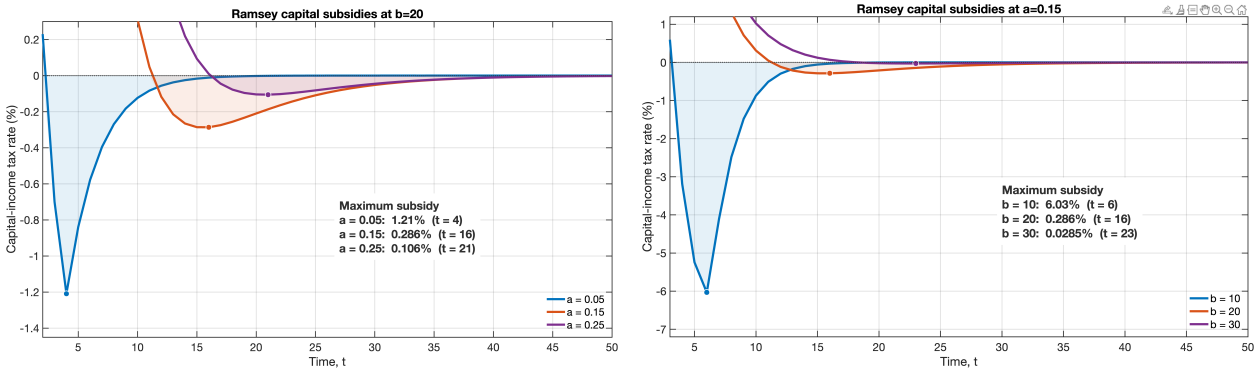


Figure 15: Ramsey capital-income subsidies during the transition. Panel (a) shows capital subsidy magnitude and duration by varying destruction exposure; Panel (b) shows the same by varying defense effectiveness. The subsidy is substantially more sensitive to defense effectiveness than to destruction exposure.

5.3 Long-run Dynamics Across Policy Regimes

The transition dynamics translate into large differences in long-run adaptation. Figure 16 reports steady-state capital, output, defense, and labor as destruction severity a varies. The first-best allocation exhibits the strong adaptation response described in Section 4.2: At low levels of destruction severity, capital falls relative to the no-destruction steady state. As destruction becomes more severe, however, the planner devotes increasing resources to defense and labor supply.

The Ramsey government reproduces part of this response. Steady state defense expenditure rises substantially with destruction severity, and capital and output recover partially after their initial decline. Nevertheless, distortionary financing prevents it from attaining the first-best allocation. In contrast, the adaptation response is substantially weaker under discretion. As destruction severity

rises, steady state defense provision increases only modestly, while capital and output fall steadily. The economy thus fails to generate the strong protection that capital accumulation needed to adapt. The differences between discretion and the other two regimes become larger as destruction becomes more severe.

The labor response also shows that adaptation is resource intensive. The first-best planner responds to greater destruction severity by mobilizing more labor together with greater defense and capital accumulation. Under discretion, labor also rises, but the much smaller capital stock and weaker protection prevent this additional labor effort from sustaining output.

Result 2 (Adaptation under Alternative Policy Regimes)**.** *For the benchmark parameterization, the first-best economy exhibits a nonmonotonic response of long-run capital and output to destruction exposure. The Ramsey government reproduces part of this adaptation, while the response is substantially weaker under discretion. The gap widens as destruction severity gets higher.*

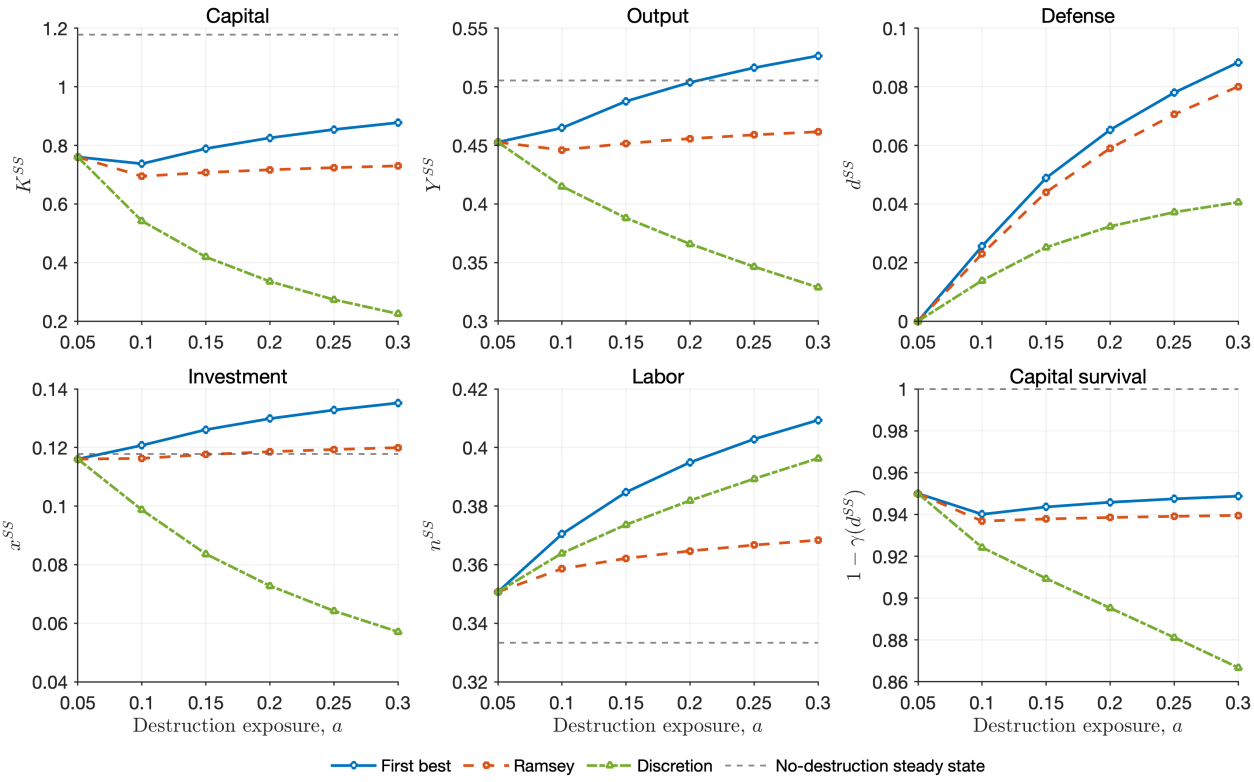


Figure 16: Steady-State Allocations across Policy Regimes as destruction exposure varies

The long-run tax structure helps explain these differences in adaptation. Figure 17 reports steady-state labor and capital income tax rates as destruction severity varies. The Ramsey government relies primarily on labor taxation and keeps the long-run capital taxation close to zero in order to preserve productive capital. Under discretion, by contrast, installed capital remains an attractive tax base and the government relies primarily on capital taxation .

As destruction becomes more severe, this difference in financing choices becomes more consequential. Under commitment, preserving investment incentives supports a large capital stock and

a stronger protection response. Under discretion, higher capital taxation depresses accumulation, leaving a smaller productive base to protect and lower protection and weakening the economy's ability to adapt.

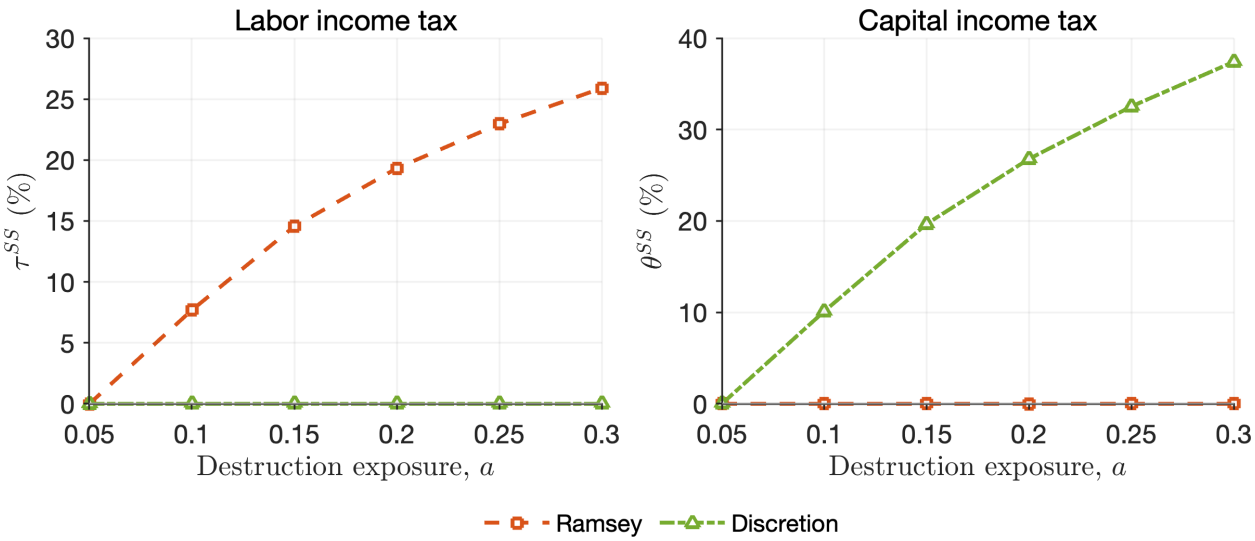


Figure 17: Tax structure under Ramsey and Discretion

This difference in tax structure matters for more than the standard capital-accumulation channel. By reducing investment and the future capital stock, capital taxation also shrinks the productive base whose survival can be improved through defense. The lack of commitment therefore weakens both productive accumulation and the protection margin.

5.3.1 Welfare and the Value of Commitment

The allocation differences under Ramsey and discretion translate into sizable welfare losses. Figure 18 reports the consumption-equivalent welfare gain from commitment relative to discretion. At the benchmark calibration, the inability to commit generates a consumption-equivalent welfare loss of 2.5 percent including the transition, and 6 percent comparing steady states, in the benchmark calibration. The welfare cost of discretion rises with destruction exposure and can reach as much as 16 percent in high stake destruction. Since lower capital accumulation also weakens the protection margin, the protection channel amplifies the welfare cost of standard time inconsistency problem.

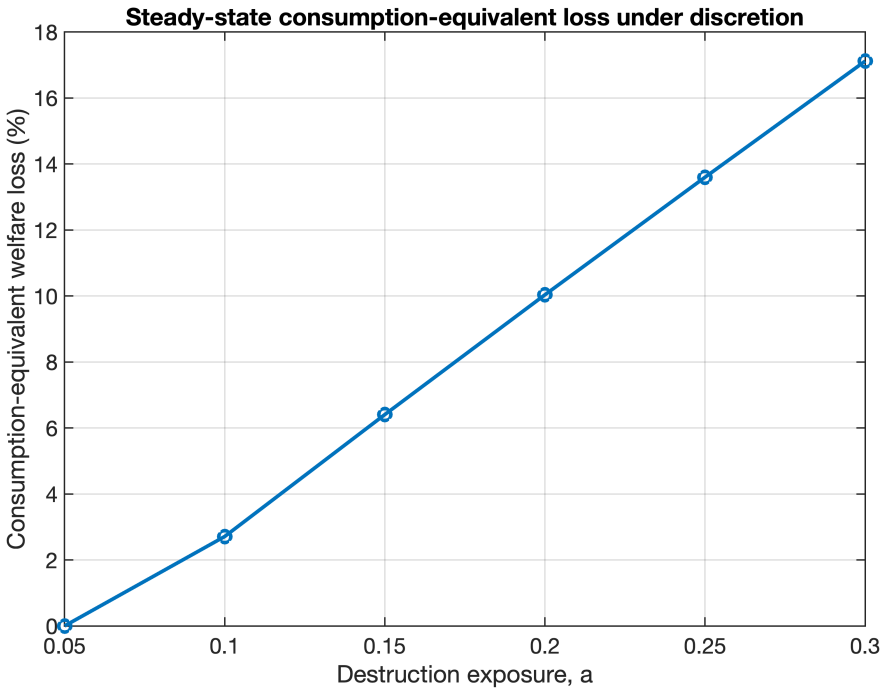


Figure 18: Welfare loss under discretion relative to Ramsey policy

Additional comparative statics. Appendix D reports additional results obtained by varying defense effectiveness b and the capital share α. These exercises assess whether the main mechanism depends on a particular benchmark technology or on the importance of capital in production.

The results are consistent with the mechanism described above. When defense is more effective, the economy can offset a larger fraction of destruction through protection, strengthening the adaptation response in capital and output. When the capital share is higher, preserving productive capital becomes more valuable. The Ramsey government therefore places greater weight on supporting capital accumulation and, for sufficiently high α, may subsidize capital during the rebuilding phase. Under discretion, the incentive to tax installed capital remains stronger, so the gap relative to Ramsey becomes more pronounced.

6 Stochastic Geopolitical Risk

The previous sections used the special case $\varphi_t = 1$ to isolate the mechanisms linking productive investment, defense, and taxation. I now return to the stochastic environment introduced in Section 3.1, in which the probability of war varies over time with geopolitical risk and destruction occurs probabilistically.

This section has two purposes. First, it connects the model more directly to the geopolitical-risk process observed in the data, by disciplining the evolution of the threat state using the Geopolitical Risk Index. Second, it studies how defense and productive investment rules vary with the current geopolitical threat state, and how distortionary financing under Ramsey alters those responses. To

isolate the role of policy rather than differences in realized shocks, I compare the first-best and Ramsey allocations under common geopolitical-threat and destruction histories.

The stochastic environment changes the interpretation of the defense decision. In the deterministic benchmark, destruction occurs with certainty, so defense directly mitigates a known loss. In the stochastic environment, defense decision is forward-looking. Defense is chosen after the current threat state is observed but before the peace or war outcome and future geopolitical-threat innovations are known. The value of preserving capital today therefore depends not only on the current probability of destruction, but also on the expected continuation value of capital across future threat states.

6.1 Calibration of Geopolitical Threat

To measure external risk faced by the economy, I use the historical threats component of the Geopolitical Risk (GPR) Index developed by Caldara and Iacoviello (2022). I construct the annual index by taking the arithmetic mean of the twelve monthly observations, and then take logs and standardize the annual series within each estimation sample.

Let z_t denote the standardized geopolitical-threat state,

$$z_t = \frac{\log(\mathrm{GPRT}_t) - \mu_{\log \mathrm{GPRT}}}{\sigma_{\log \mathrm{GPRT}}}. \tag{56}$$

Thus, $z_t = 0$ represents the average level of log geopolitical threat, while $z_t = 1$ represents a threat level one standard deviation above its historical average.

I estimate the dynamics of z_t using the AR(1) process

$$z_t = \alpha_z + \rho_z z_{t-1} + \varepsilon_t, \tag{57}$$

by ordinary least squares. The innovation variance is estimated from the OLS residuals. In particular,

$$\widehat{\varepsilon}_t = z_t - \widehat{\alpha}_z - \widehat{\rho}_z z_{t-1},$$

and the estimated innovation standard deviation is

$$\widehat{\sigma}_\varepsilon = \sqrt{\frac{\sum_{t=2}^{T} \widehat{\varepsilon}_t^2}{(T-1)-2}}.$$

Table 4 summarizes the estimated process over 1900-2019 and 1949-2019.

Mapping threat into destructive event probability. I map the threat state into the probability of a destructive event using the logistic transformation

$$\varphi_t = \Lambda(\eta_0 + z_t), \qquad \Lambda(v) = \frac{\exp(v)}{1 + \exp(v)}. \tag{58}$$

Table 4: Estimated Annual Geopolitical-Threat Process

	1900–2019	1949–2019
Mean of log annual GPRT, $\bar{g}$	4.50795	4.65150
Standard deviation of log annual GPRT, s_g	0.46711	0.27042
AR(1) intercept, $\widehat{\alpha}_z$	0.01004	-0.00218
AR(1) persistence, $\widehat{\rho}_z$	0.84478	0.69558
Innovation standard deviation, $\widehat{\sigma}_\varepsilon$	0.53148	0.72909
Annual observations	120	71

The intercept η_0 determines the probability of destructive event at the average threat level. In the benchmark calibration, I impose

$$\Pr(\text{war}_t = 1 \mid z_t = 0) = 0.30.$$

which means that average geopolitical threat corresponds to a 30% probability of a destructive event. This implies

$$\eta_0 = \Lambda^{-1}(0.30) = \log\left(\frac{0.30}{0.70}\right) = -0.84.$$

The resulting benchmark mapping is therefore

$$\varphi_t = \Lambda(-0.84 + z_t). \tag{59}$$

It implies

$$\varphi(-1) = 0.136, \qquad \varphi(0) = 0.3, \qquad \varphi(1) = 0.538.$$

Thus, the model implies that an average threat is associated with a 30% probability of destructive event, a threat one-standard-deviation below average of log GPRT is associated with a 13.6% probability of destructive event, and a threat level one standard deviation above average with a 53.8% probability.

Note that parameter η_0 is calibrated (based on assumptions) rather than estimated from destruction-onset observations. Hence, the mapping should be interpreted as a quantitative parameterization linking the observed geopolitical-threat index to the model's destructive event probability, rather than a empirical estimate of the correlation between geopolitical risk index and destruction.

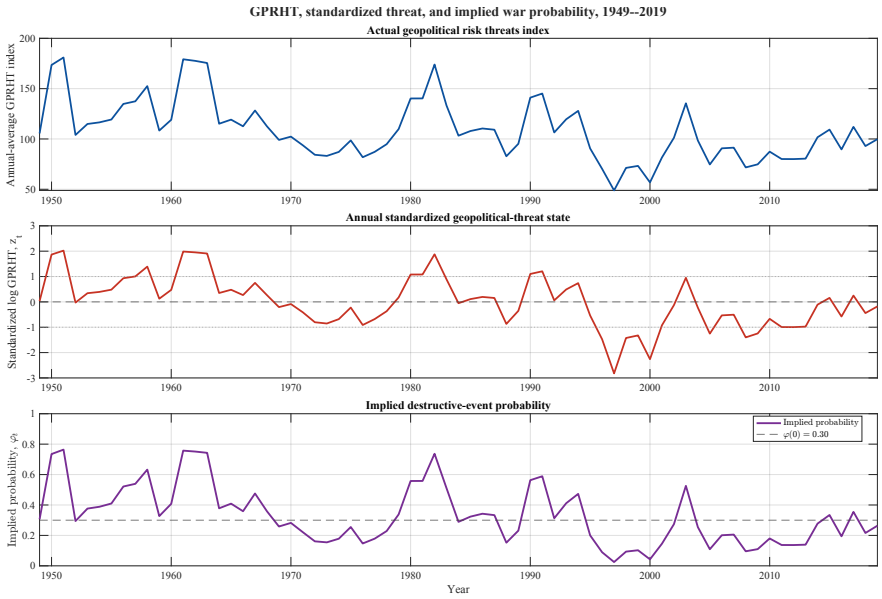


Figure 19: Geopolitical risk threats and the standardized threat state. The upper panel reports the annual-average GPRT index over 1949–2019. The middle panel reports the standardized log index. The dashed horizontal line denotes the average threat level, $z_t = 0$, and the dotted lines denote one standard deviation above and below the mean.

6.2 The stochastic first-best allocation

First best allcoations are solved with the calibrated geopolitical-threat process from 1949-2019 in Section 6.1. Figure 20 reports the resulting policy functions for productive investment and defense as functions of the current capital stock and geopolitical-threat state.

Investment is hump-shaped in capital. At low and intermediate capital levels, the planner invests more as it rebuilds the productive capital stock. At sufficiently high capital levels, investment declines. Holding capital fixed, investment generally falls as geopolitical threat increases because resources are shifted toward defense and also because capital becomes more exposed to destruction.

Defense is strongly increasing in geopolitical threat. At low threat levels, the nonnegativity constraint on defense binds over part of the capital grid. At intermediate and high threat levels, the planner chooses positive defense, and the allocation increases with both threat state and the amount of capital at risk.

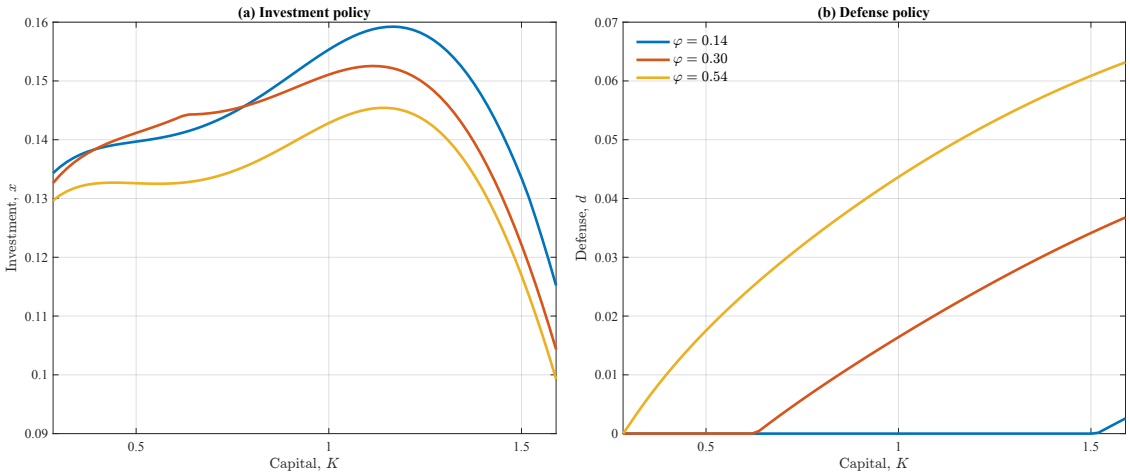


Figure 20: Stochastic first-best policy functions for investment and defense

Historical versus model-implied defense shares. Figure 21 compares the historical U.S. series with the defense allocation implied by first best model. Over 1949–2019, the average model-predicted defense share is 4.07% of output, compared with 7.45% for total defense/GDP and 3.49% for defense contracts/GDP observed in data. The corresponding contemporaneous correlations with model-predicted defense are 0.35 for total defense and 0.34 for defense contracts.

The model captures several episodes of positive comovement between geopolitical threat and defense allocations, but the model response is considerably more volatile and contemporaneous than the observed response. At first best, the planner would immediately reallocates resources toward defense when geopolitical threat rises and frequently returns to the defense nonnegativity constraint when threat falls. In contrast, observed defense spending adjusts more gradually, presumably due to budgeting, procurement, existing commitment and implementation frictions that are absent from the model.

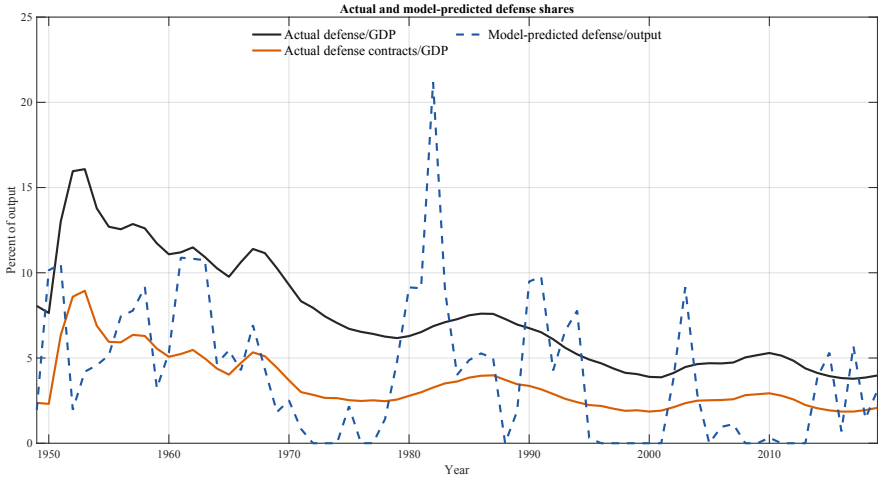


Figure 21: Actual and model-implied defense allocations, 1949–2019

Responses to a geopolitical-threat innovation I next compare the model-implied response of defense spending to a geopolitical-threat innovation with the corresponding historical response. This exercise provides a more direct quantitative test of the model's threat-response mechanism: a positive innovation to geopolitical threat should induce an increase in defense expenditure.

For the empirical response, I estimate local projections. For horizon h,

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \widehat{\varepsilon}_t + \Gamma_h X_{t-1} + u_{t+h}, \tag{60}$$

where y_t is defense contracts as a share of GDP and $\widehat{\varepsilon}_t$ is the innovation obtained from equation (**??**).

The model counterpart is a generalized impulse response. I simulate a shocked and an unshocked economy using identical subsequent innovations and compute

$$\text{GIRF}_h = E\left[\frac{d_{t+h}^{\text{shock}}}{Y_{t+h}^{\text{shock}}} - \frac{d_{t+h}^{\text{baseline}}}{Y_{t+h}^{\text{baseline}}}\middle| \varepsilon_t = \Delta_z\right]. \tag{61}$$

Figure 22 compares the empirical and model-implied responses to a one-standard-deviation geopolitical-threat innovation. Both responses are positive, consistent with the model's central mechanism that greater geopolitical threat raises the value of protection. The model, however, predicts a substantially larger and more immediate increase in defense expenditure, followed by gradual mean reversion, while the empirical response is smaller and more delayed. Since the model is in a frictionless environment, the model-implied response should be interpreted as an upper benchmark for the speed and magnitude of defense adjustment. In the data, budgeting, procurement, contractual commitments, and other implementation frictions are likely to smooth and delay the response.

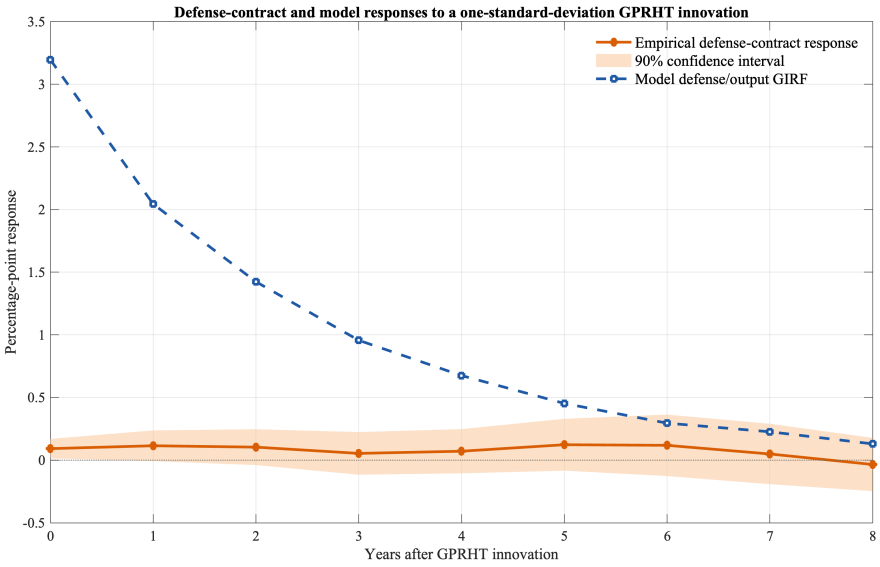


Figure 22: Empirical and model responses to a one-standard-deviation GPRT innovation. Solid lines denote empirical estimates, and shaded regions show 90% confidence intervals. Dashed blue line is model-implied defense/output response. All responses are measured in percentage points of output.

6.3 The Stochastic Ramsey Problem

The stochastic Ramsey problem introduces an inherited implementability promise, denoted by λ_t. The recursive state is therefore

$$(K_t, z_t, \lambda_t),$$

where K_t is inherited productive capital, z_t is the current geopolitical-threat state, and λ_t summarizes the implementability restriction inherited from past household saving decisions. The derivation of the state-contingent implementability condition is provided in E.

Let

$$A_t \equiv (1-\delta)K_t + x_t$$

denote exposed capital. Conditional on current choices, next-period capital is

$$K_{t+1}^P = A_t, \qquad K_{t+1}^W = S(d_t)A_t,$$

where P denotes the no-destruction state and W the destructive-event state.

The Ramsey government chooses current allocations (c_t, n_t, x_t, d_t) together with state-contingent

successor promises $\{\lambda_{t+1}^{s,j}\}_{s\in\{P,W\},\,j\in\{H,L\}}$. Its recursive problem is

$$V^R(K,z,\lambda)=\max_{\substack{c,n,x,d,\\ \{\lambda^{s,j}\}_{s,j}}}\Bigg\{u(c,n)+\beta\sum_{j\in\{H,L\}}\pi_j\Big[(1-\varphi)V^R(K^P,z^j,\lambda^{P,j}) \\ +\varphi V^R(K^W,z^j,\lambda^{W,j})\Big]\Bigg\}, \tag{62}$$

subject to feasibility,

$$c+x+d=F(K,n), \tag{63}$$

the current implementability promise,

$$\lambda=\beta\left[u_c(c,n)\big(c+x+(1-\delta)K\big)+u_n(c,n)n\right], \tag{64}$$

and the successor-promise condition,

$$u_c(c,n)A=\sum_{j\in\{H,L\}}\pi_j\left[(1-\varphi)\lambda^{P,j}+\varphi\lambda^{W,j}\right]. \tag{65}$$

The key difference from the deterministic Ramsey problem is that current saving determines only the probability-weighted average of continuation promises. The government can therefore distribute the inherited implementability requirement across the four possible successor states, (P,H), (P,L), (W,H), and (W,L).The solution algorithm is provided in Appendix F.

First best versus Ramsey under common shocks .

I first compare how the first-best and Ramsey policy rules respond to the current geopolitical-threat state. Table 5 evaluates the two allocations at a common capital stock and, for Ramsey, a common inherited implementability promise.

At low threat, both regimes choose essentially zero defense. As geopolitical threat rises, defense expenditure increases sharply. The adjustment requires a reallocation of resources: the shares of output devoted to consumption and productive investment fall, while defense absorbs a larger share of output. In the first best, labor supply also rises as the planner mobilizes additional resources.

The Ramsey allocation exhibits the same broad response, but productive investment is lower at each reported threat state. As threat rises from low to high, the Ramsey government reduces the investment share more strongly than the first-best planner. This also explains why it has a lower long-run capital stock as shown in Table 6.

Table 5: Policy responsiveness to geopolitical threat

Threat state	z	$\varphi(z)$	d/Y (%)	x/Y (%)	c/Y (%)	n
Panel A: First-best allocation						
Low GPR	-1.0	0.136	0.00	30.23	69.77	0.354
Average GPR	0.0	0.300	4.34	28.40	67.25	0.363
High GPR	1.0	0.538	9.28	26.59	64.13	0.374
Panel B: Ramsey allocation						
Low GPR	-1.0	0.136	0.00	25.42	74.58	0.345
Average GPR	0.0	0.300	5.20	23.69	71.11	0.360
High GPR	1.0	0.538	9.09	19.96	70.95	0.357

Notes: Policies are evaluated at the common no-threat capital stock $K = 1.177625$. Ramsey policies are evaluated at $\lambda = 4.876662$. The event probability is $\varphi(z) = \Lambda(-0.847298 + z)$.

I next evaluate the implications of these policy differences over stochastic histories. I let the first-best and Ramsey economies receive identical sequences of geopolitical-threat innovations and identical realizations of destructive events and simulate 5,000 paired histories. Differences between the two allocations reflect the implementability restriction. Overall, the implementability constraint produces a broad contraction in the allocation, rather than rather than a mere change in the composition of a fixed level of output. Relative to first best, the Ramsey economy has lower labor, consumption, investment, capital, and defense. Lower labor reduces current output, while lower investment reduces the future productive base. The smaller productive base, in turn, limits the resources available for both private consumption and defense.

Table 6: Long-run stochastic allocations

Variable	Mean	10th percentile	Median	90th percentile
Panel A: First-best allocation				
Consumption	0.352	0.327	0.354	0.380
Labor	0.370	0.350	0.364	0.381
Investment/output (percent)	28.534	25.946	28.466	30.806
Defense/output (percent)	4.692	0.000	3.859	10.257
Capital	1.116	0.888	1.124	1.332
Panel B: Ramsey allocation				
Consumption	0.329	0.300	0.330	0.359
Labor	0.345	0.325	0.346	0.366
Investment/output (percent)	29.051	23.357	28.719	34.867
Defense/output (percent)	4.387	0.000	3.716	10.133
Capital	1.037	0.847	1.030	1.239
Panel C: Exogenous threat state				
Threat probability	0.334	0.098	0.301	0.631

Notes: Statistics pool the final 50 periods of 5,000 paired Monte Carlo histories. Both regimes receive identical geopolitical-threat innovations and destruction-event draws. Percentiles describe the simulated long-run distribution and are not confidence intervals.

Table 7: Stochastic first-best and Ramsey allocations

Variable	First best	Ramsey	Ramsey minus first best
Consumption	0.3519	0.3290	-0.0228
Labor	0.3704	0.3452	-0.0252
Investment	0.1518	0.1434	-0.0084
Defense	0.0267	0.0217	-0.0050
Capital	1.1164	1.0373	-0.0790
Defense/output (%)	4.69	4.39	-0.31

Notes: Entries are Monte Carlo means averaged over the final 50 periods of 5,000 paired simulations. The first-best and Ramsey economies receive identical geopolitical-threat innovations and destructive-event realizations.

Figure 23 presents the complete Monte Carlo comparison. The percentile bands are wide because geopolitical shocks generate substantial aggregate uncertainty. Since the simulations are paired, both regimes face identical realizations of geopolitical-threat innovations and destructive events. Differences in their mean paths therefore reflect the how implementability constraint restricts the adjustment rather than differences in realized shocks.

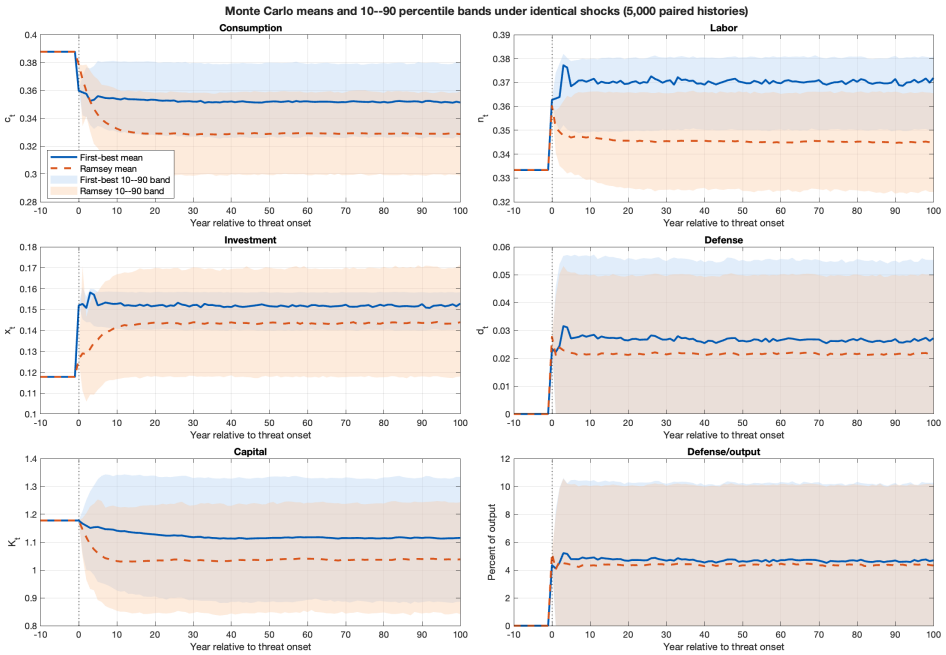


Figure 23: Stochastic first-best and Ramsey allocations under identical shocks.

7 Conclusion

This paper studies optimal defense spending and taxation in a dynamic economy in which geopolitical threats can destroy productive capital. The key feature of the environment is that productive capital is simultaneously a productive asset, the object of public protection, and a tax base that can be used to finance that protection. Defense and productive investment compete for current

resources, but they are also dynamically complementary because protection raises the destruction-adjusted return to investment, while greater investment raises the value of protection.

The model delivers three main results. First, optimal defense exhibits a threshold: when the threat is sufficiently low, or protection is sufficiently ineffective, it can be optimal to provide no defense. Second, endogenous protection creates the possibility of adaptation. When protection is sufficiently effective, greater destruction risk can induce a reallocation of resources toward both defense and productive investment, partially offsetting the decline in capital and output. This adaptation is costly, however, because it requires more defense expenditure and labor and less consumption, so welfare falls. Third, the financing of defense matters for the strength of this adaptation. A Ramsey government limits capital taxation in order to preserve investment incentives and may subsidize capital accumulation during rebuilding. Without commitment, the government instead has an incentive to tax installed capital. Anticipated taxation reduces saving and shrinks the productive base, which in turn lowers the value of protection and weakens defense. The standard capital-tax distortion is therefore amplified by the protection channel.

The stochastic version of the model, disciplined by the Geopolitical Risk Index, generates a positive response of defense spending to geopolitical-risk innovations. The response is more immediate and sharper than in the data, consistent with the model's abstraction from procurement lags and other adjustment frictions.

The analysis abstracts from several features that would be important in a richer model, including accumulated military capacity, adjustment costs in defense spending, and strategic deterrence. These extensions may affect the timing and magnitude of the defense response, but they do not remove the basic interaction between productive accumulation, protection, and fiscal policy.

The broader implication is that the fiscal consequences of destructive risk extend beyond the direct cost of protection. When the assets that require protection are also productive and taxable, the way protection is financed affects the economy's ability to preserve and rebuild its productive capacity. The framework therefore applies more broadly to environments in which productive assets face destructive risk and protection is endogenous, including natural-disaster mitigation, climate adaptation, and cybersecurity.

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A Proofs

A.1 Proof of Proposition 1

Proof. With $S(d) = 1 - ae^{-bd}$, the first-best interior condition $S(d) = AS'(d)$ becomes

$$1 - ae^{-bd} = Aabe^{-bd}.$$

Rearranging,

$$ae^{-bd}(1 + bA) = 1,$$

so an interior solution satisfies

$$d^*(A) = \frac{1}{b} \log[a(1 + bA)] .$$

Because $d \geq 0$, positive defense is optimal if and only if $a(1 + bA) > 1$, or equivalently

$$A > \frac{1 - a}{ab}.$$

Hence $A^*(a, b) = (1 - a)/(ab)$ is the threshold for positive defense. □

A.2 Proof of Proposition 2

Proof. Let

$$S^{ss}(a) = 1 - ae^{-bd^{ss}(a)}$$

and define

$$d_a^{ss} \equiv \frac{d\, d^{ss}(a)}{da}.$$

Differentiating gives

$$\frac{dS^{ss}}{da} = e^{-bd^{ss}} \left(abd_a^{ss} - 1\right). \tag{A.1}$$

Thus,

$$\frac{dS^{ss}}{da} > 0 \quad \Longleftrightarrow \quad abd_a^{ss} > 1.$$

Let $A = (1-\delta)K + x$ denote exposed capital. The interior first-best condition and the steady-state capital law are

$$S = AS_d, \qquad K = SA.$$

Because

$$S_d = abe^{-bd} = b(1-S),$$

these conditions imply

$$A = \frac{S}{b(1-S)}$$

and

$$K^{ss}(S) = \frac{S^2}{b(1-S)}. \tag{A.2}$$

Consequently,

$$\frac{dK^{ss}}{dS} = \frac{S(2-S)}{b(1-S)^2} > 0.$$

Steady-state productive investment satisfies

$$x^{ss} = A - (1-\delta)K^{ss},$$

so

$$x^{ss}(S) = \frac{S[1-(1-\delta)S]}{b(1-S)}. \tag{A.3}$$

It follows that

$$\frac{dx^{ss}}{dS} = \frac{1-(1-\delta)S(2-S)}{b(1-S)^2} > 0,$$

where the inequality uses $S(2-S) < 1$ for $S \in (0,1)$.

The steady-state Euler equation gives

$$F_K = \frac{1}{\beta S} - (1-\delta).$$

Under Cobb–Douglas production, $F_K = \alpha Y/K$. Substituting equation (A.2) therefore yields

$$Y^{ss}(S) = \frac{S[1-\beta(1-\delta)S]}{\alpha\beta b(1-S)}.$$

Hence,

$$\frac{dY^{ss}}{dS} = \frac{1-\beta(1-\delta)S(2-S)}{\alpha\beta b(1-S)^2} > 0.$$

Thus, K^{ss}, x^{ss}, and Y^{ss} are strictly increasing in effective survival. Combining this result with equation (A.1) establishes that

$$\frac{dK^{ss}}{da} > 0, \qquad \frac{dx^{ss}}{da} > 0, \qquad \frac{dY^{ss}}{da} > 0$$

if and only if

$$abd_a^{ss} > 1.$$

The corresponding derivatives are zero when $abd_a^{ss} = 1$ and negative when $abd_a^{ss} < 1$. □

A.3 Inaction region

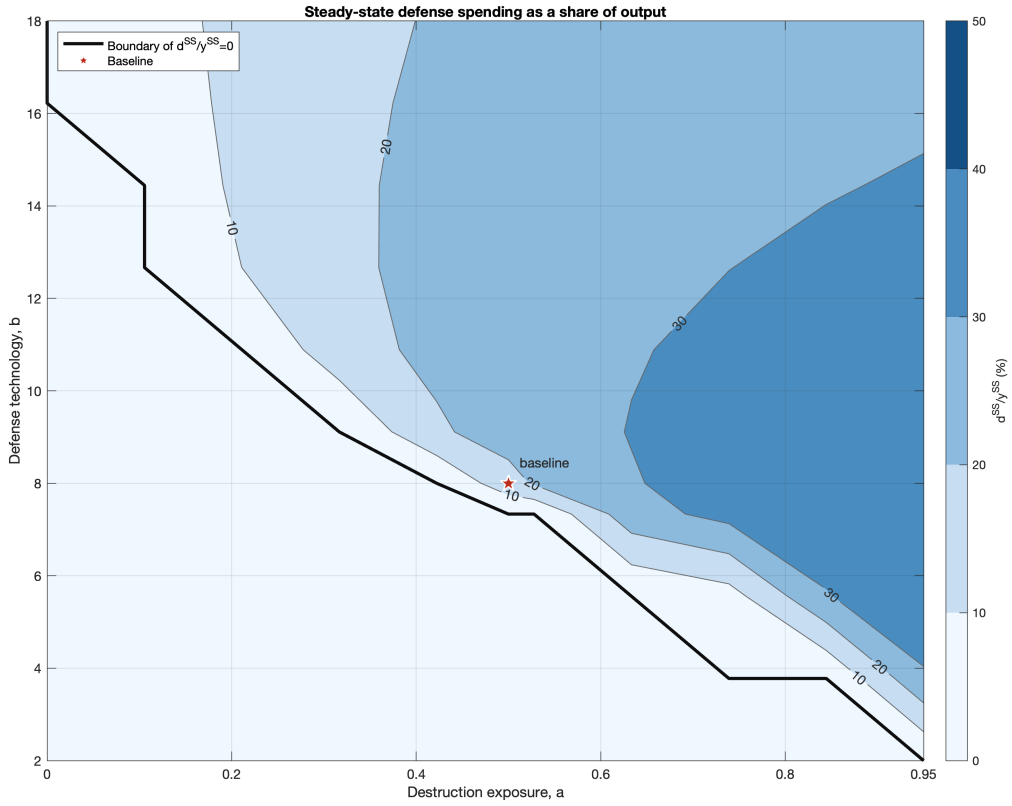


Figure 24: Defense provision

Lack of commitment changes not only how much defense is provided but also the region in which defense is optimally not provided at all.

B Alternative protection technology

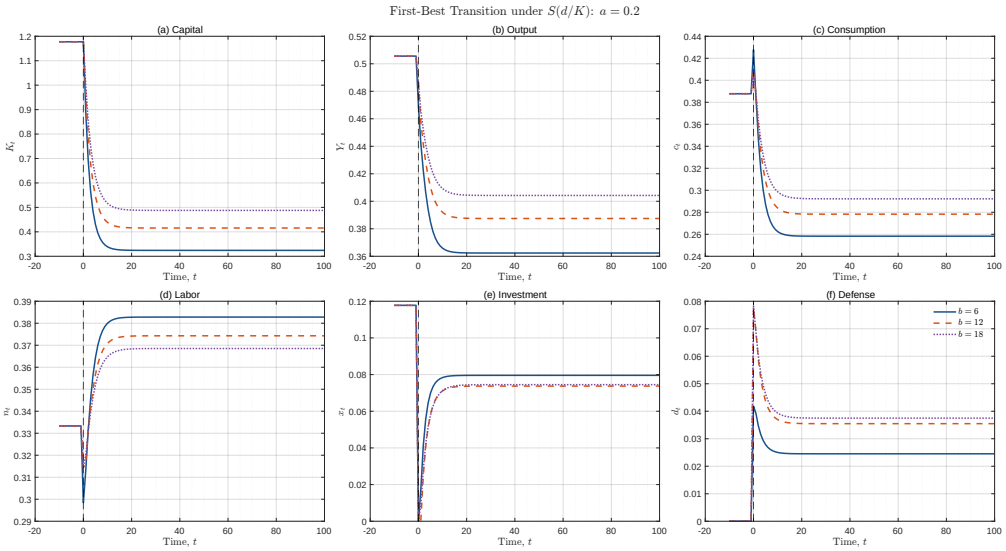


Figure 25: First best transition under $s(\frac{d}{K})$

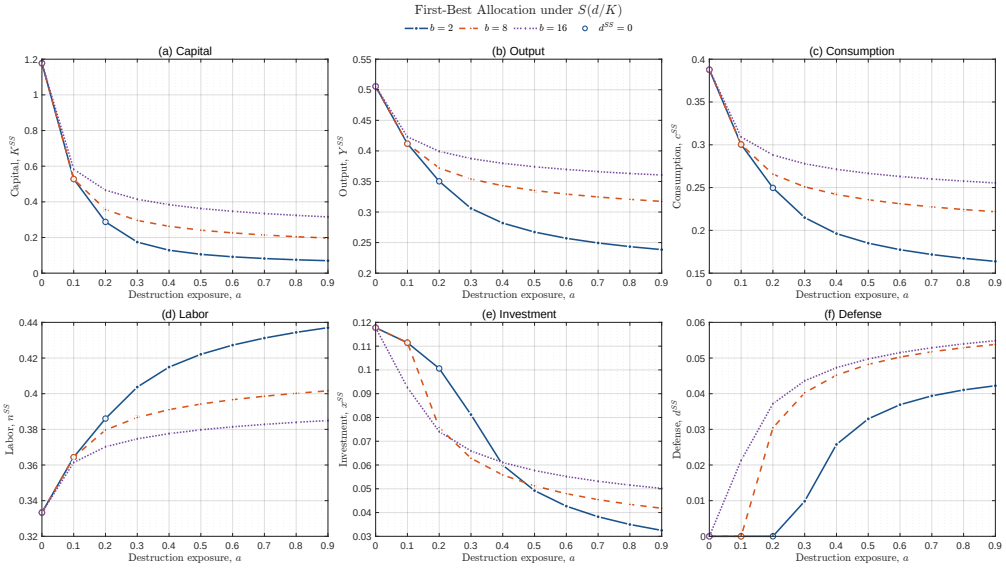


Figure 26: Steady state allocation under $S(\frac{d}{K})$, varing a

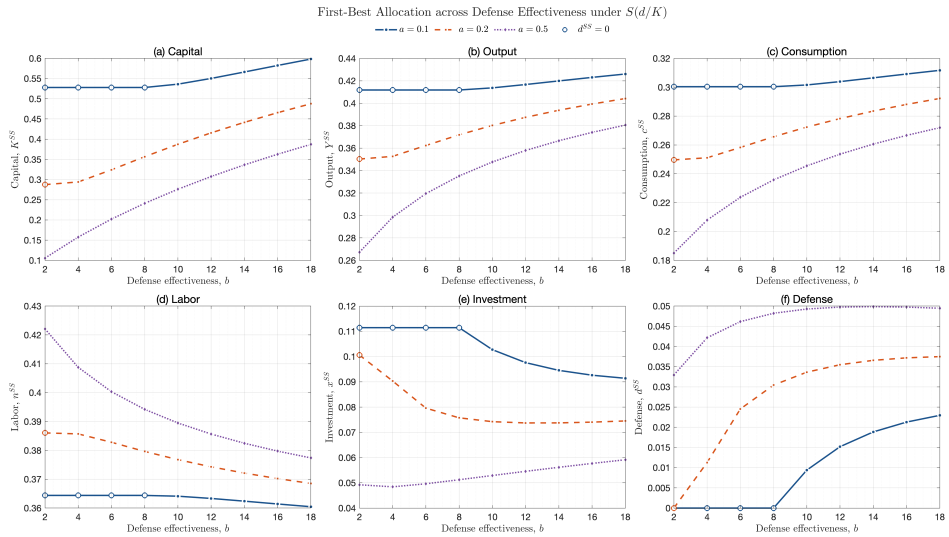


Figure 27: Steady state allocation under $S(\frac{d}{K})$, varing b

C Deterministic Ramsey Problem

This appendix provides additional details on the household problem, the derivation of the implementability constraint, the treatment of the initial period, and the recursive formulation used to solve the Ramsey problem.

C.0.1 Household Problem and Optimality Conditions

The representative household takes as given sequences of prices and government policies

$$\{w_t, r_t, \tau_t, \theta_t, d_t\}_{t=0}^{\infty}$$

and chooses

$$\{c_t, n_t, x_t, K_{t+1}\}_{t=0}^{\infty}$$

to maximize

$$\sum_{t=0}^{\infty} \beta^t u(c_t, n_t), \qquad 0 < \beta < 1, \tag{C.1}$$

subject to the household budget constraint

$$c_t + x_t = (1 - \tau_t) w_t n_t + (1 - \theta_t) r_t K_t, \qquad t \geq 0, \tag{C.2}$$

and the capital accumulation equation

$$K_{t+1} = S(d_t) \left[(1 - \delta) K_t + x_t \right]. \tag{C.3}$$

The initial capital stock K_0 is given. Competitive factor prices satisfy

$$w_t = F_n(K_t, n_t), \tag{C.4}$$
$$r_t = F_K(K_t, n_t). \tag{C.5}$$

The household's intratemporal optimality condition is

$$-u_{n,t} = u_{c,t}(1 - \tau_t)w_t, \tag{C.6}$$

where

$$u_{c,t} \equiv u_c(c_t, n_t), \qquad u_{n,t} \equiv u_n(c_t, n_t).$$

An additional unit of investment in period t increases next-period capital by

$$\frac{\partial K_{t+1}}{\partial x_t} = S(d_t).$$

One unit of capital in period $t+1$ provides the undepreciated principal $1-\delta$ together with the after-tax rental return $(1-\theta_{t+1})r_{t+1}$. The household Euler equation is therefore

$$u_{c,t} = \beta u_{c,t+1} S(d_t) \left[(1-\delta) + (1-\theta_{t+1})r_{t+1}\right]. \tag{C.7}$$

Equations (C.6) and (C.7), together with the household budget constraint, can be used to eliminate the tax rates and obtain a primal implementability constraint.

C.0.2 Derivation of the Implementability Constraint

For $t \geq 1$, lagging the household Euler equation (C.7) by one period gives

$$u_{c,t-1} = \beta u_{c,t} S(d_{t-1}) \left[(1-\delta) + (1-\theta_t)r_t\right]. \tag{C.8}$$

Therefore,

$$(1-\delta) + (1-\theta_t)r_t = \frac{u_{c,t-1}}{\beta u_{c,t} S(d_{t-1})}, \tag{C.9}$$

or equivalently,

$$(1-\theta_t)r_t = \frac{u_{c,t-1}}{\beta u_{c,t} S(d_{t-1})} - (1-\delta). \tag{C.10}$$

The labor optimality condition implies

$$(1-\tau_t)w_t n_t = -\frac{u_{n,t}}{u_{c,t}} n_t. \tag{C.11}$$

Substituting equations (C.10) and (C.11) into the household budget constraint gives

$$\begin{aligned} c_t + x_t = & -\frac{u_{n,t}}{u_{c,t}} n_t \\ & + \left[\frac{u_{c,t-1}}{\beta u_{c,t} S(d_{t-1})} - (1-\delta) \right] K_t. \end{aligned} \tag{C.12}$$

Multiplying by $\beta u_{c,t}$ and rearranging gives

$$\beta u_{c,t} \left[c_t + x_t + (1-\delta) K_t \right] + \beta u_{n,t} n_t = \frac{u_{c,t-1}}{S(d_{t-1})} K_t. \tag{C.13}$$

Using the capital accumulation equation,

$$K_t = S(d_{t-1}) \left[(1-\delta) K_{t-1} + x_{t-1} \right],$$

the right-hand side of (C.13) can be written as

$$\frac{u_{c,t-1}}{S(d_{t-1})} K_t = u_{c,t-1} \left[(1-\delta) K_{t-1} + x_{t-1} \right].$$

The implementability constraint therefore has the equivalent sequential form

$$\beta u_{c,t} \left[c_t + x_t + (1-\delta) K_t \right] + \beta u_{n,t} n_t = u_{c,t-1} \left[(1-\delta) K_{t-1} + x_{t-1} \right]. \tag{C.14}$$

To make this constraint recursive, define the inherited implementability state

$$q_t \equiv u_{c,t-1} \left[(1-\delta) K_{t-1} + x_{t-1} \right]. \tag{C.15}$$

The current implementability constraint can then be written as

$$\beta u_c(c_t, n_t) \left[c_t + x_t + (1-\delta) K_t \right] + \beta u_n(c_t, n_t) n_t = q_t. \tag{C.16}$$

Current choices determine next period's implementability state according to

$$q_{t+1} = u_c(c_t, n_t) \left[(1-\delta) K_t + x_t \right]. \tag{C.17}$$

Thus, the history relevant for current Ramsey choices can be summarized by the physical capital stock K_t and the inherited implementability state q_t.

C.0.3 Initial Period and Recovery of Tax Rates

Initial-period implementability. Because the initial capital stock K_0 is given, the treatment of the initial capital tax requires a separate condition. I rule out a nondistortionary confiscation of initial wealth by imposing

$$\theta_0 = 0. \tag{C.18}$$

The period-0 household budget constraint is therefore

$$c_0 + x_0 = (1 - \tau_0) w_0 n_0 + r_0 K_0. \tag{C.19}$$

Using the labor optimality condition,

$$(1 - \tau_0) w_0 n_0 = -\frac{u_{n,0}}{u_{c,0}} n_0,$$

gives the initial implementability condition

$$u_{c,0} \left[c_0 + x_0 - r_0 K_0 \right] + u_{n,0} n_0 = 0. \tag{C.20}$$

Using the firm's first-order condition $r_0 = F_K(K_0, n_0)$, this can equivalently be written as

$$u_{c,0} \left[c_0 + x_0 - F_K(K_0, n_0) K_0 \right] + u_{n,0} n_0 = 0. \tag{C.21}$$

Recovery of tax rates. The Ramsey problem is solved using a primal approach, so the planner chooses allocations rather than tax rates directly. Given an allocation, the tax rates that implement it can be recovered from the household optimality conditions.

From the labor condition,

$$-u_{n,t} = u_{c,t} (1 - \tau_t) F_n(K_t, n_t),$$

the labor income tax is

$$\tau_t = 1 + \frac{u_{n,t}}{u_{c,t} F_n(K_t, n_t)}. \tag{C.22}$$

From the household Euler equation,

$$u_{c,t} = \beta u_{c,t+1} S(d_t) \left[(1 - \delta) + (1 - \theta_{t+1}) F_K(K_{t+1}, n_{t+1}) \right],$$

the capital income tax satisfies

$$(1 - \theta_{t+1}) F_K(K_{t+1}, n_{t+1}) = \frac{u_{c,t}}{\beta u_{c,t+1} S(d_t)} - (1 - \delta). \tag{C.23}$$

Therefore,

$$\theta_{t+1} = 1 - \frac{\dfrac{u_{c,t}}{\beta u_{c,t+1} S(d_t)} - (1 - \delta)}{F_K(K_{t+1}, n_{t+1})}. \tag{C.24}$$

Given constant returns to scale, the household budget constraint, aggregate resource constraint, and competitive factor prices imply the government budget constraint

$$d_t = \tau_t w_t n_t + \theta_t r_t K_t.$$

C.0.4 Sequential and Recursive Ramsey Formulations

Sequential formulation. The Ramsey planner chooses

$$\{c_t, n_t, x_t, d_t, K_{t+1}\}_{t=0}^{\infty}$$

to maximize

$$\sum_{t=0}^{\infty} \beta^t u(c_t, n_t) \tag{C.25}$$

subject to the aggregate resource constraints

$$c_t + x_t + d_t = F(K_t, n_t), \qquad t \geq 0, \tag{C.26}$$

the capital accumulation equations

$$K_{t+1} = S(d_t)\left[(1-\delta)K_t + x_t\right], \qquad t \geq 0, \tag{C.27}$$

the initial implementability condition (C.20), and, for $t \geq 1$,

$$\beta u_{c,t}\left[c_t + x_t + (1-\delta)K_t\right] + \beta u_{n,t} n_t = u_{c,t-1}\left[(1-\delta)K_{t-1} + x_{t-1}\right]. \tag{C.28}$$

The initial capital stock K_0 is given.

Recursive formulation. Following the recursive approach to dynamic optimal taxation, define the implementability state q_t as in (C.15). For $t \geq 1$, the Ramsey problem can then be written recursively as

$$V^R(K, q) = \max_{c,n,x,d} \left\{u(c,n) + \beta V^R(K', q')\right\}, \tag{C.29}$$

subject to

$$c + x + d = F(K, n), \tag{C.30}$$

$$K' = S(d)\left[(1-\delta)K + x\right], \tag{C.31}$$

$$\beta u_c(c,n)\left[c + x + (1-\delta)K\right] + \beta u_n(c,n) n = q, \tag{C.32}$$

$$q' = u_c(c,n)\left[(1-\delta)K + x\right]. \tag{C.33}$$

It is useful to define

$$A \equiv (1-\delta)K + x.$$

The two state transitions can then be written compactly as

$$K' = S(d)A, \qquad q' = u_c(c,n)A.$$

The recursive state of the Ramsey economy is therefore (K, q). Physical capital K summarizes

the productive state of the economy, while q summarizes the implementability restriction inherited from past household choices.

Because the initial period has the separate implementability condition (C.20), period 0 can be written as

$$\max_{c_0,n_0,x_0,d_0} \left\{u(c_0,n_0)+\beta V^R(K_1,q_1)\right\} \tag{C.34}$$

subject to

$$c_0+x_0+d_0=F(K_0,n_0),$$

$$K_1=S(d_0)\left[(1-\delta)K_0+x_0\right],$$

$$q_1=u_{c,0}\left[(1-\delta)K_0+x_0\right],$$

and the initial implementability condition (C.20).

C.0.5 Numerical Solution

I solve the recursive Ramsey problem numerically over the state space (K,q). The numerical procedure proceeds as follows.

1. I first use long finite-horizon solutions to identify the region of the state space visited by the economy and construct buffered grids for K and q around this region.

2. For every point (K,q) on the state grid, I determine whether there exists a feasible allocation (c,n,x,d) satisfying the resource constraint and the implementability constraint

   $$\beta u_c(c,n)\left[c+x+(1-\delta)K\right]+\beta u_n(c,n)n=q.$$

   Points for which no such allocation exists are removed from the feasible state space.

3. For each feasible state, candidate controls generate next-period states

   $$K'=S(d)\left[(1-\delta)K+x\right],$$

   and

   $$q'=u_c(c,n)\left[(1-\delta)K+x\right].$$

   I require the generated state (K',q') to remain inside the state grid.

4. I solve the Bellman equation

   $$V^R(K,q)=\max_{c,n,x,d}\left\{u(c,n)+\beta V^R(K',q')\right\}$$

   over feasible controls and iterate until the value function and policy functions converge.

5. Finally, I resolve the problem using wider state grids and check that the simulated equilibrium path remains away from the grid boundaries. I also check the resource constraint, implementability constraint, and household optimality conditions along the simulated path.

D Additional Comparative Statics

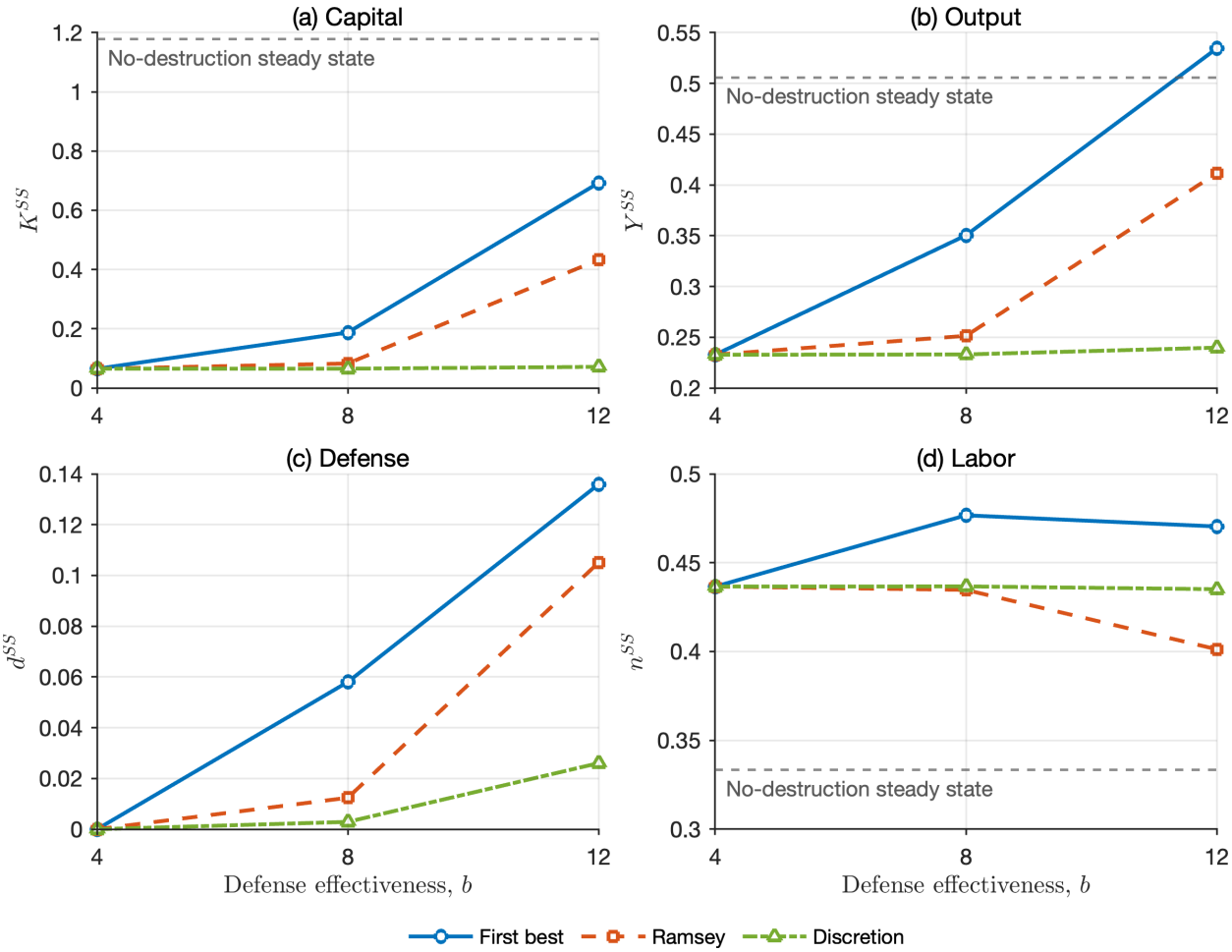


Figure 28: Productive Capacity across Defense Effectiveness

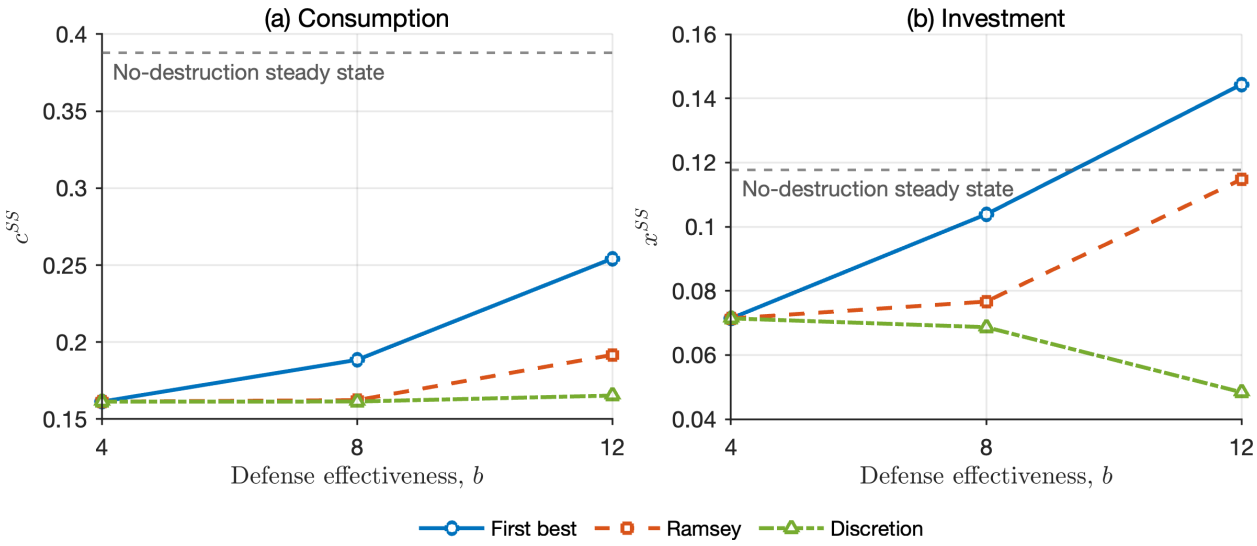


Figure 29: Consumption and Investment across Defense Effectiveness

E Stochastic Ramsey Problem

This appendix derives the implementability constraint used in the stochastic Ramsey problem. Let

$$A_t \equiv (1-\delta)K_t + x_t$$

denote capital exposed to destruction after current investment. Conditional on the current history, next-period capital is

$$K_{t+1}^{P,j} = A_t, \qquad K_{t+1}^{W,j} = S(d_t)A_t, \qquad j \in \{H, L\}.$$

For any realized successor history, the household budget constraint is

$$c_{t+1} + x_{t+1} = (1-\tau_{t+1})w_{t+1}n_{t+1} + (1-\theta_{t+1})r_{t+1}K_{t+1}. \tag{E.1}$$

The labor-supply condition implies

$$(1-\tau_{t+1})w_{t+1}n_{t+1} = -\frac{u_{n,t+1}}{u_{c,t+1}}n_{t+1}.$$

Substituting into (E.1), multiplying by $\beta u_{c,t+1}$, and adding $\beta u_{c,t+1}(1-\delta)K_{t+1}$ to both sides gives

$$\begin{aligned}&\beta u_{c,t+1}\left[c_{t+1} + x_{t+1} + (1-\delta)K_{t+1}\right] + \beta u_{n,t+1}n_{t+1} \\ &\quad = \beta u_{c,t+1}\left[1 - \delta + (1-\theta_{t+1})r_{t+1}\right]K_{t+1}.\end{aligned} \tag{E.2}$$

Taking expectations conditional on the current history and using the state-contingent capital

transition yields

$$\begin{aligned}
&\beta\mathbb{E}_t\left[u_{c,t+1}\left(c_{t+1}+x_{t+1}+(1-\delta)K_{t+1}\right)+u_{n,t+1}n_{t+1}\right] \\
&= A_t\beta\sum_{j\in\{H,L\}}\pi_j\Bigg[(1-\varphi_t)u_{c,t+1}^{P,j}\left(1-\delta+(1-\theta_{t+1}^{P,j})r_{t+1}^{P,j}\right) \\
&\qquad\qquad +\varphi_t S(d_t)u_{c,t+1}^{W,j}\left(1-\delta+(1-\theta_{t+1}^{W,j})r_{t+1}^{W,j}\right)\Bigg]. \qquad \text{(E.3)}
\end{aligned}$$

The term multiplying A_t on the right-hand side is equal to $u_{c,t}$ by the household Euler equation. Hence the stochastic implementability condition is

$$u_{c,t}A_t=\beta\mathbb{E}_t\left[u_{c,t+1}\left(c_{t+1}+x_{t+1}+(1-\delta)K_{t+1}\right)+u_{n,t+1}n_{t+1}\right]. \qquad \text{(E.4)}$$

To write the problem recursively, define

$$\lambda_t\equiv\beta\left[u_{c,t}\left(c_t+x_t+(1-\delta)K_t\right)+u_{n,t}n_t\right]. \qquad \text{(E.5)}$$

Then (E.4) becomes the promise-keeping condition

$$u_{c,t}A_t=\sum_{j\in\{H,L\}}\pi_j\left[(1-\varphi_t)\lambda_{t+1}^{P,j}+\varphi_t\lambda_{t+1}^{W,j}\right]. \qquad \text{(E.6)}$$

Thus, unlike in the deterministic problem, current saving restricts the probability-weighted average of implementability promises across successor states. The Ramsey planner can assign different continuation promises to peace and war realizations and to high- and low-threat states, subject to (E.6).

F Computational Details

The recursive state is (K,z,λ), where K is inherited productive capital, z is the geopolitical threat state, and λ is the inherited implementability promise. I solve the problem by value-function iteration.

Step 1: Construct the state space. I discretize the state space over

$$(K,z,\lambda)\in\mathcal{K}\times\mathcal{Z}\times\mathcal{L}.$$

The threat state z follows the discretized process described in Section 6. For each (K,z), I restrict the λ grid to values for which the current promise constraint is feasible.

Step 2: Solve the current-period problem. At each state (K, z, λ), the planner chooses current allocations (c, n, x, d) and four state-contingent continuation promises

$$\{\lambda^{P,H}, \lambda^{P,L}, \lambda^{W,H}, \lambda^{W,L}\}.$$

Define exposed capital as

$$A = (1-\delta)K + x.$$

The two possible next-period capital stocks are

$$K^P = A, \qquad K^W = S(d)A.$$

Current choices must satisfy the resource constraint

$$c + x + d = F(K, n),$$

the current promise constraint

$$\lambda = \beta \left[u_c(c, n) \left(c + x + (1-\delta)K \right) + u_n(c, n) n \right],$$

and the stochastic promise-keeping condition

$$u_c(c, n)A = \sum_{j \in \{H,L\}} \pi_j \left[(1 - \varphi(z))\lambda^{P,j} + \varphi(z)\lambda^{W,j} \right].$$

Step 3: Evaluate continuation values. Given a current guess $V^{(m)}$, I evaluate each feasible choice using

$$\widetilde{V} = u(c, n) + \beta \sum_{j \in \{H,L\}} \pi_j \Big[(1 - \varphi(z)) V^{(m)}(K^P, z^j, \lambda^{P,j}) + \varphi(z) V^{(m)}(K^W, z^j, \lambda^{W,j}) \Big].$$

Because $(K^P, \lambda^{P,j})$ and $(K^W, \lambda^{W,j})$ generally do not lie exactly on the grid, continuation values are obtained by interpolation over (K, λ).

Step 4: Impose continuation feasibility. A candidate choice is retained only if every continuation promise is feasible at the associated successor state:

$$\lambda^{s,j} \in \mathcal{L}(K^s, z^j), \qquad s \in \{P, W\}, \quad j \in \{H, L\}.$$

Choices generating infeasible continuation states are discarded.

Step 5: Update the value function. For each state (K, z, λ), I maximize $\widetilde{V}$ over the feasible current allocations and continuation promises and update

$$V^{(m+1)}(K, z, \lambda) = \max \widetilde{V}.$$

I iterate until the value function converges. At convergence, I store the policy functions for (c, n, x, d) and the four continuation promises $\lambda^{s,j}$.

Step 6: Solve the initial period. The initial period is treated separately because there is no inherited promise λ_0. Given (K_0, z_0), the date-zero planner chooses (c_0, n_0, x_0, d_0) and four period-one continuation promises subject to the initial implementability condition implied by $\theta_0 = 0$ and the period-one promise-keeping condition.

Step 7: Simulate the economy. Starting from the initial state, I simulate sequences of geopolitical-threat innovations and destructive-event realizations. At each date, current allocations are obtained from the policy functions, and the realized successor state selects the corresponding continuation promise. The simulations are used to compute long-run moments and to compare the Ramsey and first-best allocations under common shock histories.